How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning

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Abstract

Prompts for in-context learning (ICL) can describe a task in addition to, or instead of, demonstrating it through examples. Yet how descriptions work remains largely underexplored. Through the Bayesian view of ICL, we model descriptions as informative priors, characterized by their *specificity* and *reliability*. We adapt the effective sample size (ESS) from Bayesian statistics to quantify a description's predictive value as an equivalent number of demonstrations. Our theoretical results on in-context linear regression tasks show that a reliable description's ESS grows with its specificity and with the task's dimension. However, the ESS falls sharply if the description may be wrong: in a 16-dimensional task with no demonstrations, a one-in-ten chance of being wrong cuts a highly specific description's ESS from 117 to 13.5. We then test whether Transformer networks and pretrained language models, including frontier models, behave like Bayesian predictors in these settings. Meta-trained Transformers come close to the Bayes-optimal ESS and adapt to unreliable descriptions when trained on them. Pretrained language models, however, place too little weight on highly specific descriptions when answering without reasoning, and most of them closely match a Bayesian reference only when reasoning is enabled. Taken together, these results extend the Bayesian analysis of ICL from demonstrations to descriptions.

1. Introduction

Large language models (LLMs) are capable of in-context learning (ICL): given a prompt that specifies a task through a natural-language description, demonstrations, or both, they can predict the output for a new query input (Brown et al., 2020). In practice, prompts consist mostly of system instructions, user instructions, agent skill files, and earlier conversation turns, which typically describe the task rather than demonstrate it (Mao et al., 2025; Patsakis et al., 2026). The balance shifts further toward description as LLMs carry out increasingly long-horizon tasks (Kwa et al., 2025), for which demonstrations are costly to provide.

Theories of ICL, however, have mostly focused on prompts consisting of demonstrations alone (Xie et al., 2022; Garg et al., 2022; Akyürek et al., 2023). The few that include descriptions study only particular types, such as a task descriptor giving the mean of the inputs (Huang & Ge, 2025), a hypothesis class listed in the prompt (Lin et al., 2025), or datasets from related tasks that inform the prior (Zhu et al., 2026); none offers a systematic treatment of descriptions. We therefore ask: how can we characterize natural-language descriptions in general, so as to understand and analyze them formally? And how well do learners, whether meta-trained Transformers or pretrained language models, use descriptions in practice?

We study these questions through the Bayesian view of ICL, in which a learner infers a latent task, the function that maps each input to its output, from the demonstrations (Xie et al., 2022). A description carries information about the task without being a demonstration of it, so the description can be modeled as a prior over tasks. Descriptions differ in how much they say about a task: "write this report as an accountant" leaves most of the task open, whereas a step-by-step workflow leaves little. They also differ in whether they are about the right task: an outdated instruction, for example, describes the wrong one. We formalize these two properties in d-dimensional Gaussian linear regression, a standard controlled setting for ICL (Garg et al., 2022; Panwar et al., 2024), in which a task is a latent regression vector. A description gives a prior mean, the regression vector it points to. We call a description that is about the right task *relevant*. Given a relevant description, the task is Gaussian around its prior mean with standard deviation τ: the more a description says, the smaller τ and the higher the description's *specificity*. A description is relevant with probability p, its *reliability*; otherwise it is *irrelevant*, and

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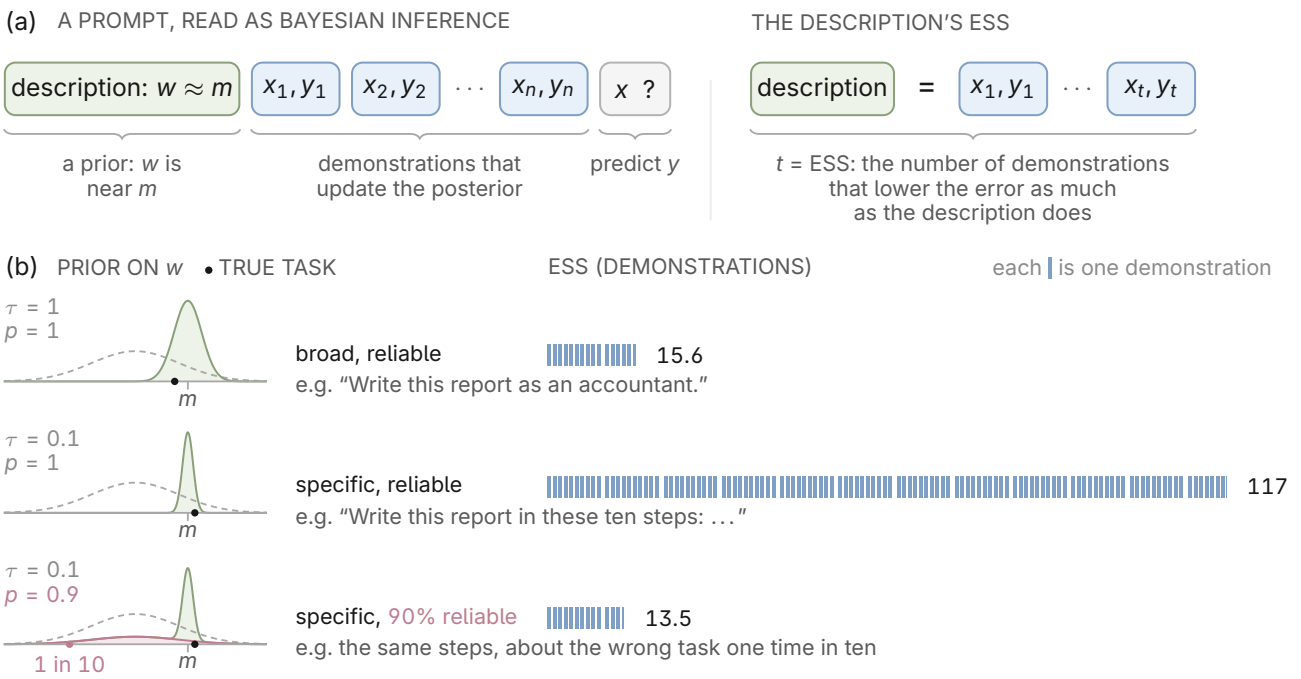


Figure 1. **Measuring a task description in demonstrations.** (a) A description's ESS is the number of demonstrations that lower the error as much as it does. (b) The prior on the task w under three descriptions, and each description's Bayes-optimal ESS at $n = 0$, with $d = 16$ and baseline prior variance $\tau_0^2 = 10$ (dashed); τ is the standard deviation of the task around the description's prior mean m. The curves are sketches; dots mark the true task, and the pink curve is the irrelevant component.

its mean is independent of the task (Figure 1).

To compare descriptions with demonstrations, we follow Bayesian work that expresses a prior's information or impact as an equivalent sample size (Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021; Selby et al., 2025), and work that expresses a prompt's value in training data points (Le Scao & Rush, 2021). A description's *predictive effective sample size* (ESS) is the number of additional demonstrations that a learner would need, in place of the description, to reach the same expected error. For the Bayes-optimal predictor, ESS gives a description's value under the statistical model; for a trained learner, it gives the value that the learner realizes. To see how well learners use descriptions, we test how close each learner comes to the ESS of a Bayesian predictor given the same information.

Our contributions are as follows:

- **Specificity and dimension determine the value of reliable descriptions.** When little is known about the task without the description, a description that places each coordinate of the task within about τ of its prior mean is worth about $c + d + 1$ demonstrations before any are seen, where $c = \tau^{-2}$ is its precision: c because each demonstration adds one unit of precision, and $d + 1$ more because random inputs leave some directions of the task poorly constrained for a learner without the description. Demonstrations lower this value, to about c after many; a description that constrains only part of the task can instead gain value from them (Section 3).
- **Unreliability caps zero-shot ESS.** Any fixed positive probability of irrelevance caps the ESS without demonstrations, the zero-shot ESS, at a finite value, however specific the description. Demonstrations can then show whether the description is relevant and raise its ESS above this ceiling (Section 4, Figure 2b).
- **Meta-trained Transformers approach the Bayes-optimal ESS when test descriptions are as reliable as training ones.** Without demonstrations, they weight a description by how often descriptions were relevant in training, whatever the test reliability (Section 5.1).
- **Language models use a precision stated in the prompt mainly when they reason.** Without reasoning, every model tested falls far short of a Bayesian predictor for what the prompt states; with reasoning, most closely match its error (Section 5.2).

2. Problem Setup

A description carries information about a task, so we model it as a prior over tasks. Let Q_0 be the *baseline prior*, the distribution of tasks when no description is given. A description comes with a prior of its own, Q_{desc}. If the description is about the right task, which we call *relevant*, the task is

drawn from Q_{desc}. Otherwise the description is *irrelevant*, and the task is drawn from Q_0 whatever the description says. Given a description, the task therefore has the prior

$$p\,Q_{\text{desc}} + (1-p)\,Q_0, \tag{1}$$

where p, the description's *reliability*, is the probability that it is relevant. We call the description *reliable* when $p = 1$ and *unreliable* when $p < 1$. This two-component form is the robust mixture prior that Bayesian clinical trials use to borrow external information that may not apply (Schmidli et al., 2014).

A description's *specificity* is how little it leaves unknown about the task when it is relevant. What a prior leaves unknown can be measured by the smallest expected loss that a prediction made from the prior alone attains, its generalized entropy (Grünwald & Dawid, 2004). We score predictions by squared error, as is standard for in-context regression (Garg et al., 2022). The best prediction from a prior is then the one made with its mean, and for the isotropic inputs used here its expected squared error is the prior's variance, summed over the directions of the task. We therefore measure a description's specificity by the variance of Q_{desc}: the lower the variance, the more specific the description. We study this model in Gaussian linear regression.

Tasks and demonstrations. A task is a vector $w \in \mathbb{R}^d$, and the baseline prior is $Q_0 = \mathcal{N}(0, \tau_0^2 I_d)$, where $\tau_0 > 0$. A prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query input x, with

$$x_i, x \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d), \qquad y_i = w^\top x_i + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

Inputs and observation noise are independent of the task, of its description, and of one another. Numerical illustrations in Sections 3–4 use $d = 16$ and $\tau_0^2 = 10$; other dimensions and baseline variances give the same pattern (Table 2 in Appendix A.2).

Descriptions. A description supplies a prior mean $m \in \mathbb{R}^d$, and its prior is $Q_{\text{desc}} = \mathcal{N}(m, \tau^2 I_d)$ with $0 < \tau \le \tau_0$. Its variance is τ^2 in each direction, so the smaller τ, the more specific the description: a relevant description places each coordinate of the task within about τ of that of m. So that learners with and without descriptions face the same tasks, the task must follow Q_0 once we average over descriptions. Under a relevant description, w is m plus independent noise of variance τ^2, so m must have variance $\tau_0^2 - \tau^2$, the part of the task's variance that the description accounts for. With a latent relevance indicator $Z \sim \text{Bernoulli}(p)$ independent of m, the joint model is

$$\begin{aligned} m &\sim \mathcal{N}(0, (\tau_0^2 - \tau^2) I_d), \\ w \mid m, Z = 1 &\sim \mathcal{N}(m, \tau^2 I_d), \\ w \mid m, Z = 0 &\sim \mathcal{N}(0, \tau_0^2 I_d). \end{aligned}$$

The description is relevant when $Z = 1$ and irrelevant when $Z = 0$, which gives the prior in (1), and integrating over m recovers $w \sim Q_0$ under either component. A description of the wrong task carries no information about this one, so when $Z = 0$ its mean is independent of the task. A description of the right task that is only somewhat inaccurate is modeled instead as a relevant description with a larger τ.

To compare a description with demonstrations, we use precision, the inverse of variance, because the precisions of independent Gaussian sources of information add. With unit noise and isotropic inputs, each demonstration adds expected precision 1 in every direction. We therefore write $c = \tau^{-2}$ for a description's precision and $c_0 = \tau_0^{-2}$ for the baseline prior's, so that both are counted in demonstrations.

Prediction error. For a prediction $\hat{y}$, we measure expected squared error against the noise-free query response:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big]. \tag{2}$$

We call this quantity *prediction error*, or simply error; it excludes irreducible query noise. The expectation is over tasks, descriptions, demonstrations, and queries. For a fixed learner, $R^{\text{desc}}(n)$ and $R^{\text{base}}(n)$ denote the error with and without a description. As a function of the demonstration count, R^{base} is the learner's *baseline curve*, its learning curve without the description.

Definition 1 (Predictive effective sample size)**.** Given n demonstrations, let n^* be the smallest $t \ge 0$ satisfying $R^{\text{base}}(t) = R^{\text{desc}}(n)$. When such a t exists, we call n^* the *crossing*, and the description's ESS is

$$\text{ESS}_n = n^* - n.$$

We interpolate the baseline curve between integer counts, so n^* and the ESS need not be integers.

For example, $\text{ESS}_n = 10$ means that n demonstrations plus the description match the error of $n + 10$ demonstrations alone. The ESS without demonstrations, ESS_0, is the *zero-shot ESS*. Definition 1 follows the prior sample size of Reimherr et al. (2021), who match the mean squared error of an estimate under a prior with that under a baseline prior and additional observations. We match prediction error, which can be measured for any learner, including a network or a language model, whose prior knowledge is only implicit in its parameters. Because ESS is measured on each learner's own baseline curve, a large ESS need not mean a small error, so we also report the language models' absolute error (Table 9, Appendix D).

3. The Value of a Reliable Description

We first take a reliable description, $p = 1$. Its value without demonstrations depends on the task's dimension as well as

on its specificity. In this section and the next, error and ESS are those of the Bayes-optimal predictor. We state the result for a diffuse baseline prior, the limit $\tau_0 \to \infty$, in which a learner knows little about the task without the description. The limit can be taken in two ways: a description may remove a fixed share of the baseline uncertainty, or its own variance τ^2 may stay fixed. For the first, let $s = 1 - \tau^2/\tau_0^2 \in [0, 1)$ be the share of the baseline variance that the description removes.

Proposition 1 (Reliable descriptions)**.** *Let* $p = 1$.

(i) With the share s *fixed,* $\mathrm{ESS}_0 \geq sd$*, and* $\mathrm{ESS}_0 \to sd$ *as* $\tau_0 \to \infty$.

(ii) With τ *fixed and* $c = \tau^{-2}$ *a positive integer,* $\mathrm{ESS}_0 \to c + d + 1$ *as* $\tau_0 \to \infty$.

Part (i) fixes the share s of the baseline variance that the description removes, so τ grows in proportion to τ_0. A learner with $t < d$ demonstrations and no description knows nothing about $d - t$ directions of w, and as $\tau_0 \to \infty$ its error is dominated by what those directions contribute, $(d - t)\tau_0^2$. The description alone leaves an error of $d\tau^2 = (1 - s)d\tau_0^2$, the variance of its prior summed over the d directions, which this learner matches at $t = sd$: a description that removes a share of the uncertainty is worth that share of d demonstrations.

Part (ii) fixes the description's variance instead. The result follows by comparing the description's error, $d\tau^2 = d/c$, with the limiting baseline error after t demonstrations, which is $d/(t - d - 1)$ for integer $t > d + 1$ (Muirhead, 1982): the two match at $t = c + d + 1$ (Appendix A.1). If the demonstration inputs gave every direction of the task the same information, the baseline error would be d/t and the description would be worth c demonstrations. Random inputs constrain some directions less than that, so the error is $d/(t - d - 1)$, and matching the description takes $d + 1$ more demonstrations.

At $d = 16$ and $\tau_0^2 = 10$ (Figure 2a), a reliable description with $\tau = 0.1$, so that $c = 100$, is worth about 117 demonstrations before any are seen, the value $c + d + 1$ that Proposition 1(ii) gives for a diffuse baseline prior.

Demonstrations lower the value of a complete description. The descriptions modeled so far are *complete*: they constrain every direction of w. After many demonstrations, the dimension no longer adds to such a description's value: $\mathrm{ESS}_n \to c - c_0$ as $n \to \infty$ (Proposition 3 in Appendix A.3). This limit is the precision that the description adds to the baseline prior, its conventional prior ESS (Morita et al., 2008) measured relative to that prior. A complete description therefore loses value as demonstrations accumulate: a $c = 100$ description is worth 117 demonstrations before any are seen and 109 after 100 (Figure 2b), and a $c = 1$ description falls from 15.6 to 1.1 (Table 3).

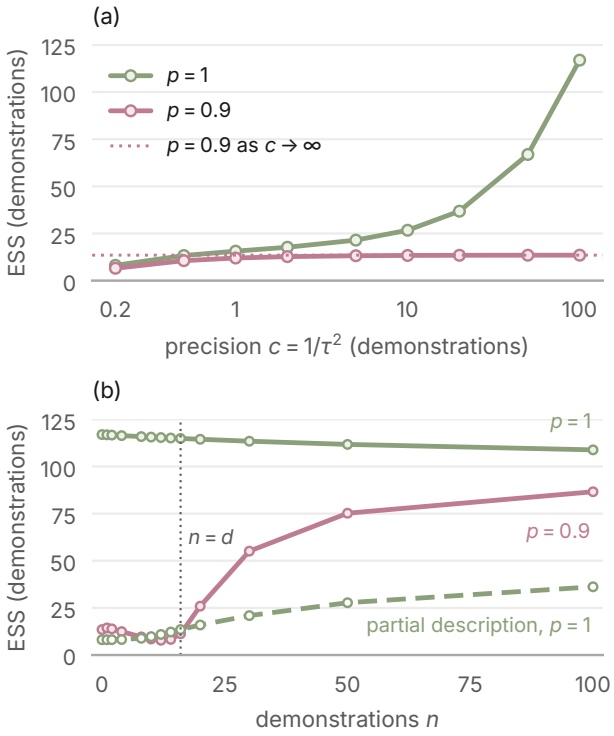


Figure 2. **Without demonstrations, reliability limits the benefit of specificity; demonstrations recover much of it.** Bayes-optimal ESS with $d = 16$ and $\tau_0^2 = 10$. (a) ESS_0 versus the description's precision $c = \tau^{-2}$, for $p = 1$ and $p = 0.9$. The dotted line is the finite ceiling as $c \to \infty$ at $p = 0.9$, about 13.5 here; it tends to $p^2 d = 12.96$ as $\tau_0 \to \infty$. At $c = 100$, that is, $\tau = 0.1$, ESS is 117 for $p = 1$ and 13.5 for $p = 0.9$. (b) ESS_n versus the number of demonstrations n for three $c = 100$ descriptions: reliable, unreliable ($p = 0.9$), and a reliable partial description, which constrains $k = 8$ of the 16 directions. Estimates use eight batches of 4,000 prompts (2,000 for the partial description); standard errors are below 0.05 demonstrations.

Demonstrations can raise the value of a partial description. Descriptions often specify only part of a task. A *partial* description constrains only $k < d$ of the d directions (Appendix A.3). Without demonstrations, its error comes almost entirely from the $d - k$ directions it leaves open, and a learner without the description reaches that error after about k demonstrations. After many demonstrations, the description is worth the precision it adds, averaged over all d directions: about $k(c - c_0)/d$ demonstrations, which exceeds k when $c - c_0 > d$. At $d = 16$ and $\tau_0^2 = 10$, a $c = 100$ partial description with $k = 8$ is worth 8.1 demonstrations before any are seen and 36.2 after 100 (Figure 2b).

4. The Cost of Unreliability

We now allow the description to be irrelevant with probability $1 - p$. The Bayes-optimal predictor must then weigh the two components by each one's probability given the demon-

strations. Let $\pi_n = P(Z = 1 \mid m, D_n)$ be the *posterior relevance probability*. The predictor averages the two components' predictions with weights π_n and $1 - \pi_n$ (Hoeting et al., 1999).

Proposition 2 (Unreliable descriptions)**.** *Let* $0 < p < 1$. *Without demonstrations,*

$$R^{\text{desc}}(0) = \left[(1-p^2)\,\tau_0^2 + p^2\,\tau^2\right] d = (1-p^2 s)\,\tau_0^2 d. \quad (3)$$

An unreliable description thus leaves the error of a reliable one that removes the share $p^2 s$ *of the baseline variance, and Proposition 1(i) holds with* $p^2 s$ *in place of* s. *Because* $p^2 s < p^2$, ESS_0 *stays below a finite ceiling for every* $\tau > 0$ *and tends to it as* $\tau \to 0$*; the ceiling is the number of demonstrations at which the baseline error equals* $(1-p^2)\tau_0^2 d$. *As* $\tau_0 \to \infty$, *the ceiling tends to* $p^2 d$, *and with* τ *fixed so does* ESS_0. *With* d, τ_0, *and* τ *fixed,* $\limsup_{n\to\infty} \text{ESS}_n \le p\,(c - c_0)$.

Without demonstrations, unreliability imposes an error floor. The error in Equation (3) is the variance of the mixture prior (1), averaged over descriptions. It is a floor, $(1-p^2)\tau_0^2 d$, plus p^2 times the error $d\tau^2$ of a reliable description. However specific the description, the error stays above the floor, and the zero-shot ESS below a finite ceiling. The floor arises because, without demonstrations, nothing shows whether the description is relevant: the posterior relevance probability is still the prior one, $\pi_0 = p$, and the Bayes-optimal predictor shrinks the prior mean m by the factor p. A predictor that knew whether the description was relevant would have a ceiling that tends to pd in place of $p^2 d$; the further factor of p is the cost of not knowing (Appendix A.2). Figure 2a shows the effect at $p = 0.9$, the weight that the robust-prior example of Schmidli et al. (2014) gives its informative component, our relevant component: lowering τ from 1 to 0.1, which raises c from 1 to 100, multiplies a reliable description's ESS by more than seven, but raises an unreliable one's only from 12.0 to 13.5, already within 0.1 of the ceiling.

Demonstrations distinguish low specificity from unreliability. By Proposition 2, an unreliable description has, without demonstrations, the error of a less specific reliable one. For example, at $d = 16$ and $\tau_0^2 = 10$, a $c = 100$ description with $p = 0.9$ and a reliable one with $c \approx 0.52$ both have ESS about 13.5. After 100 demonstrations, their ESS values are 86.6 and 0.5. Once demonstrations show that the unreliable description is relevant, its high specificity takes effect; the reliable description has low specificity to begin with, so demonstrations cannot raise its value in this way. The rise is not steady (Figure 2b, Appendix A.2), and it levels off: in the limit of many demonstrations, an unreliable description is worth at most $p(c - c_0)$, which is p times the limit $c - c_0$ of a reliable one (Proposition 2).

5. Experiments

Sections 3 and 4 give the Bayes-optimal ESS of a description. We now ask how close two kinds of learner come to it: Transformers meta-trained on our regression model, which must learn a description's specificity and reliability from their training data, and pretrained language models, whose prompts state both.

5.1. Meta-Trained Transformers

Transformers meta-trained on demonstrations can approach Bayes-optimal prediction (Garg et al., 2022; Genewein et al., 2025). We test whether they also attain the Bayes-optimal ESS of a description. Following Garg et al. (2022), we train Transformers, which we call the networks, from scratch on our regression model at $d = 5$, the dimension used by Huang & Ge (2025), with $\tau_0^2 = 10$. We train a separate network for each precision $c \in \{1, 10\}$ and each $p \in \{1, 0.99, 0.9\}$; at $c = 100$, the ESS would exceed the 20 demonstrations a network sees. A description token carries the prior mean m but not τ or p, so a network must learn both from its training data (Appendix B).

With matched reliability, the networks approach the Bayes-optimal ESS. When training and test reliability match, the networks' ESS without demonstrations is close to the Bayes-optimal ESS on the same prompts (Figure 3a). The largest gap is at $c = 10$ and $p = 1$: 14.3 against 16.1, a Bayes-optimal value close to $c + d + 1 = 16$. The networks also reproduce the gain of a partial description: with $k = 3$ of the 5 directions described, at $c = 10$ and $p = 1$, their ESS rises from 3.27 without demonstrations to 5.55 after ten, against 3.22 and 5.64 for the Bayes-optimal predictor (Appendix B).

The networks weight a description by how often descriptions were relevant in training. Without demonstrations, the Bayes-optimal predictor predicts $p\,m^\top x$, weighting the description by the test reliability p. A network instead weights it by about the reliability it was trained at (Table 4), because the description alone gives no sign of how reliable the test descriptions are. Networks trained at $p = 0.9$ thus weight descriptions by about 0.88 even when every test description is relevant, and reach only 60% of the Bayes-optimal ESS (Figure 3b). The opposite shift costs little without demonstrations: when one description in ten is irrelevant, the Bayes-optimal error is already large, and a weight of 1 in place of 0.9 adds only about 5% to it. It costs more once demonstrations arrive. For networks trained only on relevant descriptions and tested at $p = 0.9$, the description raises the error from six demonstrations onward, as it does for a Bayesian predictor that assumes relevance (Appendix B).

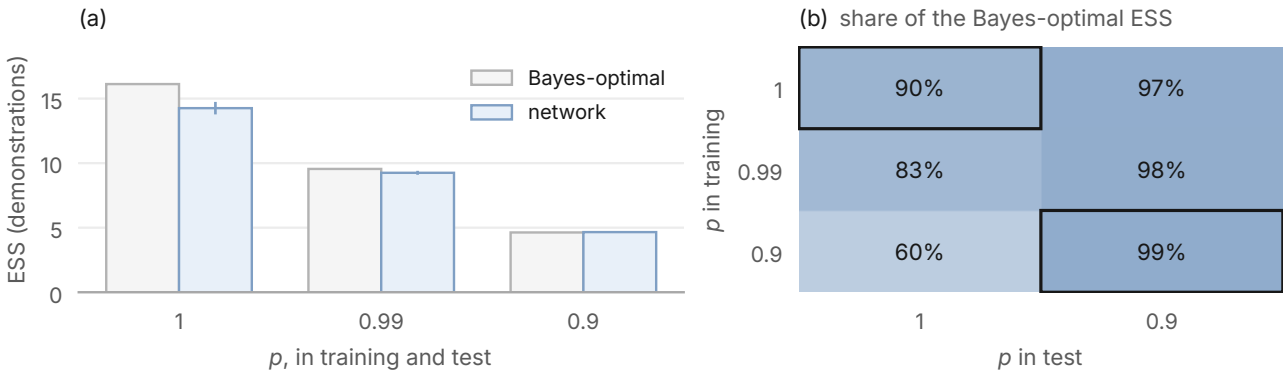


Figure 3. **Meta-trained Transformers approach Bayes-optimal ESS and weight descriptions by how often descriptions were relevant in training.** Results use $c = 10$, $d = 5$, $\tau_0^2 = 10$, and no demonstrations. (a) Training and test reliability match. Network ESS uses each network's own baseline curve; bars show the mean and one standard deviation across three training seeds. The Bayes-optimal comparison uses the same prompts. (b) Networks trained at each reliability, evaluated at each test reliability. Cells show network ESS as a fraction of the Bayes-optimal ESS under the test distribution, estimated on 200,000 prompts; outlined cells have matching reliabilities. Table 4 includes $c = 1$.

5.2. Pretrained Language Models

Pretrained language models are not trained on our task distribution, so they cannot learn a description's specificity and reliability as the networks do. We therefore state both in the prompt, and ask whether the models use what it states. To isolate the use of a description from the ability to carry out linear regression in context, we use numerical mean estimation. It is the scalar case of our regression model, with fixed input $x = 1$; with every input the same, the $d + 1$ term of Proposition 1 does not arise, and a Bayesian predictor's ESS is close to c. Using a stated precision requires weighting the description and the demonstrations by their precisions. Reasoning may help a model carry out this computation, so we compare models with reasoning enabled and disabled.

Task and prompts. Each task has a hidden mean $w \sim \mathcal{N}(500, 100^2)$, and its demonstrations are drawn from $\mathcal{N}(w, \sigma^2)$ with $\sigma = 60$, which the prompt states. A description gives a prior mean m and a standard deviation τ: "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30." We generate m as in Section 2, so the task lies within about τ of it. Each demonstration now adds precision σ^{-2}, so the description's precision counted in demonstrations is $c = \sigma^2/\tau^2$; stated standard deviations of 60, 30, and 10 give $c = 1$, 4, and 36. Because a more precise description also has a more accurate prior mean, a model could do well with $c = 36$ descriptions without using the stated precision. To separate the two, we also give the $c = 36$ descriptions with a stated precision of $c = 4$ (Appendix C.1).

Models and reference. We give the same 200 tasks to five models whose reasoning can be enabled and disabled: gpt-oss-20b (OpenAI, 2025), Qwen3.5 9B (Qwen Team, 2026), Gemma 4 31B (Gemma Team, 2026), GPT-6 Sol (OpenAI, 2026), and Claude Opus 4.6 (Anthropic, 2026b). Claude Haiku 5.5 (Anthropic, 2026a), whose reasoning could not be disabled in our setup, appears only with reasoning. We compare each model with a *Bayesian reference*, the Bayesian predictor for the statistical model that the prompt states. With a description, the reference predicts approximately the precision-weighted mean $(cm + n\bar{y})/(c + n)$, where $\bar{y}$ is the sample mean of the demonstrations. Without one, the prompt says only that the mean is "unknown", because a stated distribution would itself be a description, and the reference takes a uniform prior over the integers 100 to 999, the range of every number shown. This prior is not the normal distribution that generates the tasks, so agreement with the reference shows that a model follows what the prompt states, not that its error is the lowest possible on these tasks. Models rarely answer with a number when the prompt has neither a description nor demonstrations, so we compute ESS at $n = 1$, 2, and 4 against each model's own baseline curve and report the median of the three values, the *median ESS*; the reference's is 1.0, 3.5, and 34.6 at $c = 1$, 4, and 36 (Appendix C.2).

Language models use a stated precision mainly when they reason. Without reasoning, all five models fall far short of the reference at $c = 36$: median ESS ranges from 2.0 to 10.3, against 34.6 for the reference (Figure 4b). With reasoning enabled, four of the six models come close to the reference's absolute error at $c = 4$ and $c = 36$: Opus 4.6, Gemma 4 31B, and Qwen3.5 9B are within 0.1 of it at every demonstration count, and GPT-6 Sol within 0.3 at all but one (Table 9). Opus 4.6, GPT-6 Sol, and Gemma 4 31B also attain its ESS at both precisions (Figure 4a,b), and when only the stated precision is lowered, from $c = 36$ to $c = 4$, their ESS falls to between 11 and 14, against 13.7 for the ref-

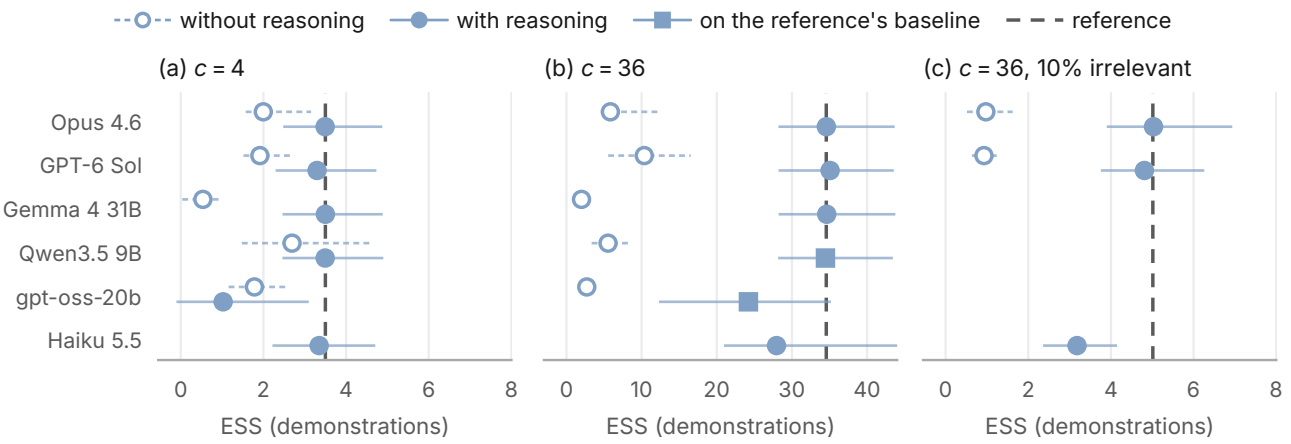


Figure 4. **Language models use the precision and the reliability that a prompt states mainly when they reason.** Median ESS over $n = 1$, 2, and 4, from error averaged over 200 tasks at $\sigma = 60$; 95% bootstrap intervals resample tasks. Open points indicate reasoning disabled; filled points indicate reasoning enabled; vertical dashed lines show the Bayesian reference. (a) Descriptions generated and stated at $c = 4$. (b) Descriptions generated and stated at $c = 36$. (c) GPT-6 Sol, Opus 4.6, and Haiku 5.5 on the $c = 36$ descriptions of (b), when the prompt states a 10% chance that the description is irrelevant; each task is presented with a relevant description and with an irrelevant one. Squares in (b) measure ESS on the reference's baseline curve, not the model's own, and so are not comparable with the other points. Intervals are clipped to the display.

erence (Table 8), so they use the stated precision itself and do not merely average the demonstrations better. For some models, the runs with and without reasoning also differ in serving software and retry policy, so these are comparisons of the configurations we evaluated, not evidence for a particular mechanism (Appendix C.4, which also covers the other two models and the squares in Figure 4b).

Stated reliability. Section 4 shows that demonstrations can reveal whether an unreliable description is relevant. To test whether language models use demonstrations in this way, we add a sentence to the $c = 36$ description: "There is a 10% chance that this statement is wrong, in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999." Each task is presented once with its relevant description and once with an irrelevant one, and we average the two errors with weights 0.9 and 0.1. For cost reasons, we ran this experiment on three of the models, GPT-6 Sol, Opus 4.6, and Haiku 5.5 (Appendix C.6). The stated chance cuts the reference's ESS from 34.6 to 5.0 (Figure 4c). Most of what remains comes from using the demonstrations to judge whether the description is relevant: a predictor that keeps the chance of irrelevance at the stated 10%, whatever the demonstrations show, has an ESS of only 0.3. With reasoning, GPT-6 Sol and Opus 4.6 reach 4.8 and 5.0, and Haiku 5.5 reaches 3.2; without reasoning, GPT-6 Sol and Opus 4.6 reach only 0.9 and 1.0 (Table 10).

Descriptions in words. The descriptions above state a distribution, whereas descriptions in practice usually state constraints in words. We therefore ran two of these three models, GPT-6 Sol and Haiku 5.5, on a second task, message sorting (Appendix C.8). A hidden rule, one of 60, sorts messages into two folders by two of their six properties, and the description is a note that narrows down the rule, such as "to go into A, a message must request a payment". A note leaves the rules compatible with it equally likely, so it is more specific the fewer rules it leaves: 20 for a broad note and 2 for a specific one (Appendix D relates this count to the measure of Section 2). Models give a probability for each message's folder, scored by squared error, and the reference is the exact Bayesian predictor over the 60 rules. With reasoning, both models' ESS is within 2 demonstrations of the reference's for every note and demonstration count but one; a specific note at $n = 2$, for example, is worth 9.5 and 9.9 demonstrations to them, against 11.4 for the reference (Table 13). When the prompt adds that the note has a 50% chance of being wrong, 77 to 99% of their predictions are closer to the reference than to a predictor that keeps that chance fixed, at $n = 2$ and 4.

6. Related Work

Theories of ICL. Prior work studies ICL as Bayesian inference over latent tasks (Xie et al., 2022; Panwar et al., 2024), as a learning algorithm implemented in a forward pass (Akyürek et al., 2023; von Oswald et al., 2023), and as meta-learning across tasks (Garg et al., 2022; Müller et al., 2022). Meta-trained predictors can approximate Bayesian inference under suitable training conditions (Genewein et al., 2025). Agreement with Bayesian predictors can depend on training-task diversity and on the function class (Raventós

et al., 2023; Panwar et al., 2024), and language-model ICL can violate the martingale property that Bayesian prediction implies (Falck et al., 2024). Lin & Lee (2024) study a prior that mixes task groups, a mixture like the one an unreliable description induces here. In these accounts, the prior over tasks comes from pretraining or meta-training. The works closest to ours put particular kinds of prior information in the prompt (Huang & Ge, 2025; Lin et al., 2025; Zhu et al., 2026); we characterize a description by two properties that any description has, its specificity and its reliability, and measure its value in demonstrations.

Prior effective sample size. Prior ESS expresses the information or impact of a prior as an equivalent sample size, with different definitions matching different statistical quantities (Morita et al., 2008; Neuenschwander et al., 2020). Our measure is closest to those based on expected loss (Reimherr et al., 2021; Wiesenfarth & Calderazzo, 2020). Selby et al. (2025) compare priors elicited from language models with data-based predictors; their measure is the number of observations needed to match predictive loss. We apply loss matching to task descriptions, at each demonstration count and against each learner's own baseline curve. Related measures express a prompt or an instruction in training examples: Le Scao & Rush (2021) measure a prompt's average data advantage in fine-tuning, and Cook et al. (2026) find that, in fine-tuning, an instruction can replace up to 100 examples. We measure descriptions used in context.

Descriptions and evidence in language models. Task summaries distilled from demonstrations can replace the examples in context (Honda et al., 2025). Other studies report weak sensitivity to instruction relevance (Webson & Pavlick, 2022) or limited benefits from repository-level context files (Gloaguen et al., 2026). Gupta et al. (2025) give language models an explicitly stated coin bias together with observed flips, evaluate their predictions against Bayesian predictors, and find that in the models' predictions the observed flips outweigh the stated bias. Tong et al. (2026) study how sensitivity to the prompt template varies with the number of demonstrations. Our language-model experiments differ in stating the precision and the reliability of a description separately, measuring its value in demonstrations against each model's own baseline curve, and comparing reasoning enabled and disabled. Concurrent work studies how reasoning and code execution affect language models' approximation of Bayesian posteriors (Filbry et al., 2026); we vary only reasoning, with code execution disabled. Appendix D discusses further connections.

7. Discussion

We asked how descriptions can be characterized and how well learners use them in practice. For the first question, we treated a description as a prior with a specificity and a reliability, and we measured it by its predictive ESS. A reliable description can replace many demonstrations; unreliability imposes a finite ESS ceiling before demonstrations are observed; and demonstrations can then raise a description's ESS by supplying evidence about its relevance. For the second, meta-trained Transformers approach the Bayes-optimal ESS and, without demonstrations, weight a description by how often descriptions in their training data were relevant. Language models make too little use of a precise description without reasoning; with reasoning, most use a stated precision as the Bayesian reference does.

Our controlled analysis suggests why a description can substitute for many examples (Honda et al., 2025) or have limited influence (Webson & Pavlick, 2022; Gupta et al., 2025; Gloaguen et al., 2026). A description has little influence when it says little about the task, when it may be about the wrong task, when the demonstrations already supply what it says, or when the learner does not use it. The Bayes-optimal ESS reflects the first three, and a learner's shortfall from it reflects the fourth. For prompt design, the results suggest two practices. An instruction that may be wrong should come with enough demonstrations to let the learner check it, and a prompt that warns of this should say what holds in that case. In a variant of our stated-reliability prompt that said only that the mean was otherwise unknown, the models we tested revised the stated chance far less than the Bayesian reference; once the prompt said what held instead, the two models we retested revised it as the reference does (Appendix C.6). More broadly, treating a description as a prior extends the Bayesian analysis of ICL from demonstrations to descriptions.

Limitations and future work. We model a relevant description as a single Gaussian prior around one prior mean. This is the usual case: a description points to one task and is more or less specific about it. Future work could model other forms of description, such as one that names several candidate tasks, with a mixture of Gaussians, which can approximate a wide class of priors (Dalal & Hall, 1983). Most of our language-model prompts state a description's precision and reliability explicitly, as numbers, so that every prediction can be compared with a Bayesian reference. In practice, these properties are implicit: a more specific instruction is typically a longer one that says more about the task. The message-sorting notes are a first step in this direction. How well models infer these properties from the wording of realistic instructions is the next question, and predictive ESS applies to it directly, since it requires only a learner's errors with and without the description. Our language-model experiments also use small synthetic tasks, and two of them cover only two or three models.

Impact Statement

This work studies how learners combine task descriptions with demonstrations. Its measurement framework can inform prompt evaluation and design when descriptions might be about the wrong task or leave much of it open. The experiments use synthetic tasks; their results do not establish dependable instruction following in consequential applications.

References

Abdin, M., Aneja, J., Behl, H., et al. Phi-4 technical report. Technical report, Microsoft, 2024. arXiv:2412.08905.

Akata, E., Voudouris, K., Fortuin, V., and Schulz, E. In-context function learning in large language models. In *AISTATS*, 2026.

Akyürek, E., Schuurmans, D., Andreas, J., Ma, T., and Zhou, D. What learning algorithm is in-context learning? Investigations with linear models. In *ICLR*, 2023.

Anthropic. System card: Claude Haiku 5.5. https://www.anthropic.com/claude-haiku-5-5-system-card, 2026a.

Anthropic. System card: Claude Opus 4.6. https://anthropic.com/claude-opus-4-6-system-card, 2026b.

Arora, A., Jurafsky, D., Potts, C., and Goodman, N. D. Bayesian scaling laws for in-context learning. In *COLM*, 2025.

Brown, T. B. et al. Language models are few-shot learners. In *NeurIPS*, 2020.

Cook, J., Sapora, S., Ahmadian, A., Khan, A., Rocktäschel, T., Foerster, J., and Ruis, L. Programming by backprop: an instruction is worth 100 examples when finetuning LLMs. In *ICLR*, 2026.

Dalal, S. R. and Hall, W. J. Approximating priors by mixtures of natural conjugate priors. *Journal of the Royal Statistical Society: Series B (Methodological)*, 45(2):278–286, 1983.

Falck, F., Wang, Z., and Holmes, C. Is in-context learning in large language models Bayesian? A martingale perspective. In *ICML*, 2024.

Filbry, L., Melo, L. C., and Gal, Y. From prompts to posteriors: Can LLM agents perform Bayesian inference in context? *arXiv:2610.29331*, 2026.

Garg, S., Tsipras, D., Liang, P., and Valiant, G. What can Transformers learn in-context? A case study of simple function classes. In *NeurIPS*, 2022.

Gemma Team. Gemma 4 technical report. *arXiv:2607.02770*, 2026.

Genewein, T., Wenliang, L. K., Grau-Moya, J., Ruoss, A., Orseau, L., and Hutter, M. Understanding prompt tuning and in-context learning via meta-learning. In *NeurIPS*, 2025.

Gloaguen, T., Mündler-Sasahara, N., Müller, M. N., Raychev, V., and Vechev, M. Evaluating AGENTS.md: are repository-level context files helpful for coding agents? *arXiv:2602.11988*, 2026.

Goodman, N. D., Tenenbaum, J. B., Feldman, J., and Griffiths, T. L. A rational analysis of rule-based concept learning. *Cognitive Science*, 32(1):108–154, 2008.

Grünwald, P. D. and Dawid, A. P. Game theory, maximum entropy, minimum discrepancy and robust Bayesian decision theory. *The Annals of Statistics*, 32(4):1367–1433, 2004.

Gruver, N., Finzi, M., Qiu, S., and Wilson, A. G. Large language models are zero-shot time series forecasters. In *NeurIPS*, 2023.

Gupta, R., Corona, R., Ge, J., Wang, E., Klein, D., Darrell, T., and Chan, D. M. Enough coin flips can make LLMs act Bayesian. In *ACL*, 2025.

He, Z., Qiu, T., Shirado, H., and Sap, M. Martingale Score: An unsupervised metric for Bayesian rationality in LLM reasoning. In *NeurIPS*, 2025.

Hoeting, J. A., Madigan, D., Raftery, A. E., and Volinsky, C. T. Bayesian model averaging: A tutorial. *Statistical Science*, 14(4):382–417, 1999. doi: 10.1214/ss/1009212519.

Honda, U., Murakami, S., and Zhang, P. Distilling many-shot in-context learning into a cheat sheet. In *Findings of EMNLP*, 2025.

Huang, R. and Ge, R. Task descriptors help Transformers learn linear models in-context. In *ICLR*, 2025.

Kwa, T., West, B., Becker, J., Deng, A., Garcia, K., Hasin, M., Jawhar, S., Kinniment, M., Rush, N., Von Arx, S., et al. Measuring AI ability to complete long software tasks. *arXiv:2503.14499*, 2025.

Le Scao, T. and Rush, A. M. How many data points is a prompt worth? In *NAACL*, 2021.

Lin, Z. and Lee, K. Dual operating modes of in-context learning. In *ICML*, 2024.

Lin, Z., Bharti, S. K., and Lee, K. In-context learning with hypothesis-class guidance. *arXiv:2502.19787*, 2025.

Mao, Y., He, J., and Chen, C. From prompts to templates: A systematic prompt template analysis for real-world LLMapps. In *FSE Industry Track*, 2025. arXiv:2504.02052.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *EMNLP*, 2022.

MiniCPM Team. MiniCPM5-2B. https://huggingface.co/openbmb/MiniCPM5-2B, 2026. Model card; the series is described in *MiniCPM4: Ultra-Efficient LLMs on End Devices*, arXiv:2506.07900.

Morita, S., Thall, P. F., and Müller, P. Determining the effective sample size of a parametric prior. *Biometrics*, 64(2):595–602, 2008.

Muirhead, R. J. *Aspects of Multivariate Statistical Theory*. Wiley, 1982.

Müller, S., Hollmann, N., Pineda Arango, S., Grabocka, J., and Hutter, F. Transformers can do Bayesian inference. In *ICLR*, 2022.

Neuenschwander, B., Weber, S., Schmidli, H., and O'Hagan, A. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020.

OpenAI. gpt-oss-120b & gpt-oss-20b model card. Technical report, OpenAI, 2025. arXiv:2508.10925.

OpenAI. GPT-6 Sol and GPT-6 Luna: October 2026 update. https://deploymentsafety.openai.com/gpt-6-october, 2026.

Panwar, M., Ahuja, K., and Goyal, N. In-context learning through the Bayesian prism. In *ICLR*, 2024.

Patsakis, C., Argyropoulos, V., and Alepis, E. Configuration, not conscience: A large-scale empirical study of LLM system prompts. In *ICTAI*, 2026. arXiv:2609.31575.

Qiu, L., Sha, F., Allen, K., Kim, Y., Linzen, T., and van Steenkiste, S. Bayesian teaching enables probabilistic reasoning in large language models. *Nature Communications*, 17:1238, 2026.

Qwen Team. Qwen3.5-9B. https://huggingface.co/Qwen/Qwen3.5-9B, 2026. Model card.

Raventós, A., Paul, M., Chen, F., and Ganguli, S. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *NeurIPS*, 2023.

Reimherr, M., Meng, X.-L., and Nicolae, D. L. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(3):413–437, 2021.

Requeima, J., Bronskill, J., Choi, D., Turner, R. E., and Duvenaud, D. LLM processes: numerical predictive distributions conditioned on natural language. In *NeurIPS*, 2024.

Schmidli, H., Gsteiger, S., Roychoudhury, S., O'Hagan, A., Spiegelhalter, D., and Neuenschwander, B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.

Selby, D., Iwashita, Y., Spriestersbach, K., Saad, M., Bappert, D., Warrier, A., Mukherjee, S., Kise, K., and Vollmer, S. Had enough of experts? Quantitative knowledge retrieval from large language models. *Stat*, 14: e70054, 2025.

Team Olmo. Olmo 3. *arXiv preprint arXiv:2512.13961*, 2025.

Team OLMo, Walsh, P., Soldaini, L., Groeneveld, D., Lo, K., et al. 2 OLMo 2 Furious. In *COLM*, 2025. arXiv:2501.00656.

Tong, X., Zeng, Y., and Zhang, J. Demonstrations, CoT, and prompting: a theoretical analysis of ICL. *arXiv:2603.19611*, 2026.

von Oswald, J., Niklasson, E., Randazzo, E., Sacramento, J., Mordvintsev, A., Zhmoginov, A., and Vladymyrov, M. Transformers learn in-context by gradient descent. In *ICML*, 2023.

von Rosen, D. Moments for the inverted Wishart distribution. *Scandinavian Journal of Statistics*, 15(2):97–109, 1988.

Webson, A. and Pavlick, E. Do prompt-based models really understand the meaning of their prompts? In *NAACL*, 2022.

Wei, J., Wei, J., Tay, Y., Tran, D., Webson, A., Lu, Y., Chen, X., Liu, H., Huang, D., Zhou, D., and Ma, T. Larger language models do in-context learning differently. *arXiv:2303.03846*, 2023.

Weller, O., Lourie, N., Gardner, M., and Peters, M. E. Learning from task descriptions. In *EMNLP*, pp. 1361–1375, 2020. doi: 10.18653/v1/2020.emnlp-main.105.

Wenliang, L. K., Ruoss, A., Grau-Moya, J., Hutter, M., and Genewein, T. Why is prompting hard? Understanding prompts on binary sequence predictors. In *AISTATS*, 2026. PMLR 300:2134–2142.

Wiesenfarth, M. and Calderazzo, S. Quantification of prior impact in terms of effective current sample size. *Biometrics*, 76(1):326–336, 2020.

Xie, S. M., Raghunathan, A., Liang, P., and Ma, T. An explanation of in-context learning as implicit Bayesian inference. In *ICLR*, 2022.

Zhu, Q., Oermann, E. K., and Cho, K. Multi-task Bayesian in-context learning. In *ICML*, 2026. arXiv:2606.20538.

A. Proofs

A.1. Reliable Descriptions

Notation and error. As in Section 2, $c_0 = 1/\tau_0^2$ is the baseline prior precision and $c = 1/\tau^2$ is the description's. Let $X_n \in \mathbb{R}^{n\times d}$ contain the demonstration inputs as rows, and set $W_n = X_n^\top X_n$. For a prior precision $\kappa > 0$, define

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I_d + W_n)^{-1}.$$

Thus $R^{\mathrm{base}} = R^{c_0}$ and, for a reliable description, $R^{\mathrm{desc}} = R^c$. Here and throughout, error is the expected squared prediction error against the noise-free query response $w^\top x$. Both identities hold because the Bayes-optimal prediction is the posterior mean. Given the prompt and a query x, its expected squared error is $x^\top \Sigma x$, where $\Sigma = (\kappa I_d + W_n)^{-1}$ is the posterior covariance; averaging over the independent query, with $\mathbb{E}[xx^\top] = I_d$, gives $\mathrm{tr}\,\Sigma$.

Throughout the proofs, d is a positive integer, $0 < \tau \le \tau_0$, so that $c \ge c_0$, $s = 1 - \tau^2/\tau_0^2 \in [0, 1)$, and $p \in [0, 1]$. Specificity is measured by the relevant component's variance τ^2; s is the share of the baseline variance that the description removes. Reliability p is the prior probability of relevance. Noninteger demonstration counts refer to linear interpolation between the errors at adjacent integers.

Wishart identities. We use three standard properties of W_n.

(W1) If $n < d$, then W_n has rank n almost surely. Consequently, $(\kappa I_d + W_n)^{-1}$ has $d - n$ eigenvalues equal to $1/\kappa$.

(W2) If $n \ge d$ and e is a fixed unit vector, then $(e^\top W_n^{-1} e)^{-1} \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12). Hence

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = \frac{d}{n-d-1} \quad (n \ge d+2),$$

whereas this expectation is infinite for $n = d$ and $n = d + 1$.

(W3) For fixed d, as $n \to \infty$,

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-2} = \frac{d}{n^2} + O(n^{-3}), \qquad \mathbb{E}\,\mathrm{tr}\,W_n^{-3} = O(n^{-3})$$

(von Rosen, 1988).

The Sherman–Morrison identity implies that an additional demonstration strictly decreases the expected trace of the posterior covariance. Thus $R^\kappa(n)$ is strictly decreasing, and (W2) implies that it tends to zero. Its linear interpolation is therefore continuous and strictly decreasing. In the correctly specified joint model, conditioning on the description cannot increase expected posterior variance, so $R^{\mathrm{desc}}(n) \le R^{\mathrm{base}}(n)$ and the Bayes-optimal ESS is nonnegative.

Diffuse baseline prior. Several results use the baseline error measured in units of the baseline variance τ_0^2. For every integer n,

$$\frac{R^{c_0}(n)}{\tau_0^2} = \mathbb{E}\,\mathrm{tr}(I_d + \tau_0^2 W_n)^{-1} \longrightarrow (d-n)^+ \qquad \text{as } \tau_0 \to \infty.$$

Indeed, an eigenvalue $1/(1 + \tau_0^2\lambda)$ is bounded by one and converges to the indicator of $\lambda = 0$, so dominated convergence and the almost-sure rank of W_n apply. The same limit holds at every real $t \ge 0$ by interpolation. Each finite-τ_0 curve is strictly decreasing, and monotone functions that converge pointwise to a continuous limit converge locally uniformly. Hence, if an error level $\ell(\tau_0)$ satisfies $\ell(\tau_0)/\tau_0^2 \to \lambda$ with $0 < \lambda < d$, the count at which the baseline error equals $\ell(\tau_0)$ converges to $d - \lambda$, the only point where $(d - t)^+ = \lambda$.

Proof of Proposition 1. A reliable description without demonstrations has error $R^{\mathrm{desc}}(0) = d\tau^2 = (1 - s)d\tau_0^2$, the trace of its prior covariance.

(i) Fixed share. By (W1), $R^{c_0}(n) \ge (d - n)\tau_0^2$ for $n < d$; the inequality also holds trivially for $n \ge d$. Linear interpolation preserves this bound, so

$$R^{\mathrm{base}}(t) \ge (d - t)\tau_0^2 \qquad (t \ge 0).$$

For $t < sd$, the right-hand side exceeds $(1-s)d\tau_0^2$. The crossing n^* of Definition 1 therefore satisfies $n^* \geq sd$, so $\mathrm{ESS}_0 \geq sd$. In units of τ_0^2, the description error is $(1-s)d$. For $s \in (0,1)$, this level lies strictly between 0 and d, so by the diffuse-baseline limit the crossing converges to $d-(1-s)d = sd$. For $s=0$, the description is uninformative and $\mathrm{ESS}_0 = 0$ for every τ_0.

(ii) Fixed specificity. For fixed τ, the description error $d\tau^2 = d/c$ does not depend on τ_0. As $\tau_0 \to \infty$, monotone convergence, (W1), and (W2) give

$$R^{c_0}(n) \longrightarrow \begin{cases} +\infty, & n \leq d+1, \\ \dfrac{d}{n-d-1}, & n \geq d+2. \end{cases}$$

The finite part of this limiting curve equals d/c at the integer $n = c+d+1$. For every $t < c+d+1$ the limiting error exceeds d/c, and for every $t > c+d+1$ it is smaller, so the crossing of the interpolated finite-τ_0 error converges to $c+d+1$. The baseline curve is interpolated linearly between integer counts, so the limit is exactly $c+d+1$ only for integer c, and the proposition is stated for that case. □

A.2. Unreliable Descriptions

The posterior relevance probability is $\pi_n = P(Z=1 \mid m, D_n)$, whereas p remains the prior relevance probability. Conditional on $Z=1$, the posterior parameters are

$$\Sigma_1 = (cI_d + W_n)^{-1}, \qquad \mu_1 = \Sigma_1(cm + X_n^\top y_{1:n}).$$

Conditional on $Z=0$, the description's mean m is independent of w, and

$$\Sigma_0 = (c_0 I_d + W_n)^{-1}, \qquad \mu_0 = \Sigma_0 X_n^\top y_{1:n}.$$

Because m has the same distribution under both relevance states, its density cancels from the posterior odds, and the posterior relevance probability follows from the two Gaussian marginal likelihoods of the demonstrations,

$$\begin{aligned} L_1 &= \mathcal{N}\big(y_{1:n}; X_n m, I_n + \tau^2 X_n X_n^\top\big), \\ L_0 &= \mathcal{N}\big(y_{1:n}; 0, I_n + \tau_0^2 X_n X_n^\top\big), \\ \pi_n &= \frac{pL_1}{pL_1 + (1-p)L_0}. \end{aligned}$$

The query input does not enter this update because it is independent of (Z, m, D_n). Without demonstrations, both marginal likelihoods equal one, so $\pi_0 = p$. The Bayes-optimal prediction is the posterior mean, $\big(\pi_n\mu_1 + (1-\pi_n)\mu_0\big)^\top x$ (Hoeting et al., 1999).

Proof of Proposition 2. At $n=0$, $\mu_1 = m$, $\mu_0 = 0$, and $\pi_0 = p$, so the prediction is $p\,m^\top x$. For each coordinate,

$$\mathrm{Var}(w_i) = \tau_0^2, \qquad \mathrm{Var}(m_i) = \tau_0^2 - \tau^2 = s\tau_0^2, \qquad \mathrm{Cov}(w_i, m_i) = ps\tau_0^2.$$

It follows that

$$\mathbb{E}(w_i - pm_i)^2 = \tau_0^2 - 2p^2 s\tau_0^2 + p^2 s\tau_0^2 = (1-p^2 s)\tau_0^2 = (1-p^2)\tau_0^2 + p^2\tau^2.$$

Since $\mathbb{E}[xx^\top] = I_d$, the prediction error is $\mathbb{E}\|w - pm\|^2$. Summing over coordinates proves Equation (3). Because $p\,m$ is the mean of the mixture prior (1) given m, this error is the trace of that prior's covariance, averaged over m.

Equation (3) has the form $(1-s')d\tau_0^2$ with $s' = p^2 s$, the zero-shot error of a reliable description that removes the share s'. The proof of Proposition 1(i) uses the description only through this error, so it applies with $p^2 s$ in place of s: $\mathrm{ESS}_0 \geq p^2 sd$, and $\mathrm{ESS}_0 \to p^2 sd$ as $\tau_0 \to \infty$ with s fixed.

Since $R^{\mathrm{base}}(0) = d\tau_0^2$ and $s < 1$, Equation (3) gives, for every $\tau > 0$,

$$R^{\mathrm{desc}}(0) = (1-p^2 s)R^{\mathrm{base}}(0) > (1-p^2)R^{\mathrm{base}}(0).$$

The error on the left decreases to the right-hand side as $\tau \to 0$, equivalently $s \to 1$. The interpolated baseline error is continuous, strictly decreasing, and tends to zero. Let $n_p^\sharp$ be the count at which the baseline error equals $(1-p^2)\tau_0^2 d$, the

ceiling of the proposition; it is unique and finite. The crossing n^* is strictly below $n_p^\sharp$ and converges to it as $\tau \to 0$. Finally, the error level that defines $n_p^\sharp$ is $(1-p^2)d$ in units of τ_0^2, which lies strictly between 0 and d. By the diffuse-baseline limit of Appendix A.1, $n_p^\sharp \to d-(1-p^2)d = p^2 d$ as $\tau_0 \to \infty$. With τ fixed instead of s, $s \to 1$ as $\tau_0 \to \infty$, and the same limit gives $\mathrm{ESS}_0 \to p^2 d$.

For comparison, a predictor that also observes Z predicts $m^\top x$ when $Z=1$ and 0 when $Z=0$, with zero-shot error

$$p\,d\tau^2 + (1-p)\,d\tau_0^2 = (1-ps)\,d\tau_0^2.$$

As $\tau \to 0$ this error tends to $(1-p)d\tau_0^2$, so by the same limit its zero-shot ESS ceiling approaches pd as $\tau_0 \to \infty$, against $p^2 d$ for the Bayes-optimal predictor, which does not observe Z. The difference, $p(1-p)d$, is the cost of not knowing whether the description is relevant.

Many demonstrations. The same predictor bounds the ESS after many demonstrations. Given Z, it is the Bayes-optimal predictor for the corresponding component, so its error with n demonstrations is $pR^c(n) + (1-p)R^{c_0}(n)$. It predicts with the posterior mean given (Z, m, D_n), and the Bayes-optimal predictor with the posterior mean given (m, D_n) alone. Conditioning on less cannot lower the expected squared error, so

$$R^{\mathrm{desc}}(n) \geq pR^c(n) + (1-p)R^{c_0}(n).$$

The expansion in the proof of Proposition 3(ii), applied with $\Lambda = cI_d$ and with $\Lambda = c_0 I_d$, gives for every fixed real $t \geq 0$

$$R^{\mathrm{base}}(n+t) - \left[pR^c(n) + (1-p)R^{c_0}(n)\right] = \frac{d}{n^2}\left[p\,(c-c_0) - t + O(n^{-1})\right].$$

Fix $t > p(c-c_0)$. For all large n the right-hand side is negative, so $R^{\mathrm{base}}(n+t) < R^{\mathrm{desc}}(n)$. The crossing n^* exists, because $R^{\mathrm{desc}}(n) \leq R^{\mathrm{base}}(n)$ and the interpolated baseline error is continuous and tends to zero (Appendix A.1); since that error is strictly decreasing, $n^* < n+t$, that is, $\mathrm{ESS}_n < t$. Since $t > p(c-c_0)$ is arbitrary, $\limsup_{n\to\infty} \mathrm{ESS}_n \leq p(c-c_0)$: even a predictor that knew whether the description was relevant would gain the added precision $c-c_0$ only with probability p. □

Simulations indicate that the bound is nearly attained in the settings we study. The ESS can stay below it in the limit because the Bayes-optimal predictor remains unsure whether the description is relevant, which the predictor that observes Z is not. Table 1 gives Monte Carlo estimates of this gap (`results/theory-check/vn_limit.py`): it is below 10^{-4} demonstrations at $d=16$ and grows as the dimension falls. The bound concerns the limit, not each finite n: at $d=16$, $\tau_0^2 = 10$, $c=100$, and $p=0.9$, where the bound is 89.91, the ESS is 86.6 at $n=100$, 90.6 at $n=400$, and 90.1 at $n=3{,}000$, so it approaches the limit from above.

Table 1. **The bound of Proposition 2 after many demonstrations is nearly attained.** The bound $p(c-c_0)$ on an unreliable description's ESS in the limit $n \to \infty$, and a Monte Carlo estimate of how far the ESS stays below it, at $\tau_0^2 = 10$. The gap is estimated from the difference in error between the Bayes-optimal predictor and the predictor that observes Z, with two million draws.

d	c	p	bound $p(c-c_0)$	gap
16	100	0.9	89.91	< 0.0001
16	1	0.9	0.81	< 0.0001
5	10	0.9	8.91	0.002
5	1	0.9	0.81	0.010
5	1	0.5	0.45	0.023
1	36	0.9	32.31	1.30
1	1	0.9	0.81	0.105
1	1	0.5	0.45	0.199

Other dimensions and baseline variances. The numerical illustrations in the body use $d=16$ and $\tau_0^2 = 10$. Table 2 repeats the comparison of a reliable and an unreliable $c=100$ description at other values. In every case the reliable description's zero-shot ESS is within 1.6 of $c+d+1$. A one-in-ten chance of irrelevance cuts it to between 4.2 and 44.4, and it approaches the diffuse-prior limit $p^2 d$ of Proposition 2 as τ_0^2 grows.

Table 2. **The cost of unreliability at other dimensions and baseline variances.** Bayes-optimal zero-shot ESS of a $c = 100$ description at $p = 1$ and $p = 0.9$, with the limits of Propositions 1 and 2. The baseline curve is estimated from 4,000 sets of demonstration inputs per count (1,500 at $d = 50$).

d	τ_0^2	ESS_0, $p = 1$	$c + d + 1$	ESS_0, $p = 0.9$	$p^2 d$
5	1	105.0	106	8.7	4.05
5	10	105.8	106	4.7	4.05
5	100	105.9	106	4.2	4.05
16	1	115.8	117	17.3	12.96
16	10	116.9	117	13.5	12.96
16	100	117.0	117	13.0	12.96
50	1	149.5	151	44.4	40.50
50	10	150.9	151	40.9	40.50
50	100	151.0	151	40.5	40.50

ESS with demonstrations. The unreliable description's ESS in Figure 2b does not rise steadily: after a small rise at $n = 1$, from 13.5 to 14.4, it falls to 7.8 at $n = 12$, and it climbs steeply only as n approaches d, to 11.3 at $n = 16$ and 25.9 at $n = 20$. With fewer demonstrations than d, its error is dominated by the cases in which it is irrelevant. The baseline curve falls steeply while directions of the task remain undetermined, so it matches that error within a few demonstrations.

A.3. Partial Descriptions

Descriptions often specify only part of a task. In our model, a reliable description may then constrain only k of the d directions of w; Proposition 3 gives its ESS, and the case $k = d$ is the complete description of Section 3. Formally, in the k constrained coordinates, $m_i \sim \mathcal{N}(0, \tau_0^2 - \tau^2)$ and $w_i \mid m \sim \mathcal{N}(m_i, \tau^2)$, as in Section 2; in the remaining $d - k$ coordinates, $m_i = 0$ and $w_i \sim \mathcal{N}(0, \tau_0^2)$. The prior precision given the description is $c = 1/\tau^2$ in the constrained directions and $c_0 = 1/\tau_0^2$ elsewhere. Write

$$\Lambda = \mathrm{diag}(cI_k, c_0 I_{d-k}).$$

The prediction error is $\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1}$. Rotational invariance of W_n means that the result depends on the number of constrained directions, not on their orientation.

Proposition 3 (ESS of a partial description)**.** *Let $p = 1$ and let $0 \le k \le d$ be an integer.*

(i) If $k < d$, then $\mathrm{ESS}_0 \to k$ as $\tau_0 \to \infty$ with τ fixed.
(ii) For fixed d, τ_0, and τ, as $n \to \infty$,

$$\mathrm{ESS}_n \longrightarrow \frac{k}{d}(c - c_0) = \frac{k}{d}\, sc.$$

In particular, a complete description has $\mathrm{ESS}_n \to c - c_0$.

As the baseline prior becomes diffuse with τ fixed, the unconstrained directions dominate the zero-shot error, and ESS approaches the number of constrained directions, k. With even one unconstrained direction, the error with the description stays of order τ_0^2, which a learner without the description matches after about k demonstrations; only a complete description brings the error down to order τ^2, which takes $c + d + 1$ demonstrations to match. After many demonstrations, ESS instead approaches the added precision averaged across all d directions. When $c - c_0 > d$, this large-sample ESS, $k(c - c_0)/d$, exceeds the diffuse-prior zero-shot ESS k, so for a diffuse enough baseline prior the description is worth more after many demonstrations than before any. Table 3 illustrates this: at $c = 100$, a description of half of the $d = 16$ directions has ESS 8.1 without demonstrations and 36.2 with 100 demonstrations.

Proof of Proposition 3. (i) No demonstrations. The description error is $\mathrm{tr}\,\Lambda^{-1} = (d - k)\tau_0^2 + k\tau^2$. In units of τ_0^2, it is

$$d - k + \frac{k\tau^2}{\tau_0^2} \longrightarrow d - k \qquad \text{as } \tau_0 \to \infty.$$

If $0 < k < d$, this limit lies strictly between 0 and d, so by the diffuse-baseline limit of Appendix A.1 the crossing converges to $d - (d - k) = k$. If $k = 0$, the description leaves the prior unchanged and its ESS is zero.

Table 3. **Demonstrations can increase the value of a specific partial description.** Bayes-optimal ESS for a reliable description of k of $d = 16$ directions, with $\tau_0^2 = 10$ and precision $c = \tau^{-2}$. The case $k = 16$ is a complete description. Each block reports ESS after n demonstrations and its large-sample limit $(c - c_0)k/d$. Estimates use eight batches of 2,000 prompts; standard errors are at most 0.05.

	$c = 1$					$c = 100$				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

(ii) Many demonstrations. Applying the resolvent identity twice gives

$$(\Lambda + W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda + W_n)^{-1}.$$

Rotational invariance implies

$$\mathbb{E}W_n^{-2} = \frac{\mathbb{E}\,\mathrm{tr}\,W_n^{-2}}{d} I_d.$$

The expected trace of the middle term is therefore $\mathrm{tr}(\Lambda)/n^2 + O(n^{-3})$ by (W3). To bound the remainder, use $|\,\mathrm{tr}(MB)| \le \|M\|_{\mathrm{op}}\,\mathrm{tr}\,B$ for $B \succeq 0$, together with $(\Lambda + W_n)^{-1} \preceq W_n^{-1}$. Its absolute trace is at most

$$\|\Lambda\|_{\mathrm{op}}^2 \|W_n^{-1}\|_{\mathrm{op}}^2\,\mathrm{tr}\,W_n^{-1} \le d\|\Lambda\|_{\mathrm{op}}^2\,\mathrm{tr}\,W_n^{-3},$$

whose expectation is $O(n^{-3})$. Consequently,

$$\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n - d - 1} - \frac{\mathrm{tr}\,\Lambda}{n^2} + O(n^{-3}).$$

The baseline error is the case $\Lambda = c_0 I_d$; if $k = 0$ or $c = c_0$, the two errors coincide and $\mathrm{ESS}_n = 0$. Otherwise, fix an integer $T > k(c - c_0)/d$. Since $d/(n + t - d - 1) = d/(n - d - 1) - dt/n^2 + O(n^{-3})$, uniformly over integers $0 \le t \le T$,

$$R^{\mathrm{base}}(n + t) - \mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n^2}\left[\frac{\mathrm{tr}\,\Lambda - dc_0}{d} - t + O(n^{-1})\right].$$

Linear interpolation gives the same expansion for real $t \in [0, T]$. The baseline error is strictly decreasing, so for large n the left-hand side vanishes at exactly one t in this range, which is ESS_n. Hence

$$\mathrm{ESS}_n = \frac{\mathrm{tr}\,\Lambda - dc_0}{d} + O(n^{-1}) \longrightarrow \frac{k(c - c_0)}{d}.$$

One more demonstration and one more unit of prior precision in every direction both lower the error by d/n^2 to leading order, which is why a description's large-sample ESS is the precision it adds, averaged across directions. □

B. Transformer Experiments

Setup. Following Garg et al. (2022), we train from scratch on the model of Section 2, with $\tau_0^2 = 10$ and $d = 5$, the dimension used by Huang & Ge (2025). A prompt contains an optional description token carrying m, up to 20 demonstration tokens, and a query token (Huang & Ge, 2025). The description token does not state τ or p; a network must learn these properties across training tasks. Half the prompts omit the description; a network's predictions on such prompts give its own baseline curve. The main networks are trained separately for $c \in \{1, 10\}$ and $p \in \{1, 0.99, 0.9\}$; further networks, described later in this section, cover $c = 0.2$ and 2, $d = 10$, and a partial description. The two main precisions are a factor of ten apart, as are the two nonzero probabilities of irrelevance, $1 - p = 0.01$ and 0.1, so that a few networks span a wide range.

Matched training and test reliability. When training and test reliability match, the networks come close to the Bayes-optimal ESS (Figure 3a; Table 4). The largest gap is at $c = 10$ and $p = 1$, where networks reach a mean ESS of 14.3 against 16.1 on the same prompts (Table 4 gives 15.9, from a larger sample); in every other configuration, they are within 0.3 of the Bayes-optimal ESS.

Reliability shifts. Each network learns how much to weight a description from how often descriptions were relevant in its training data, but descriptions at test time may be more or less reliable than that. We therefore evaluate each network at test reliabilities that differ from its training reliability, and compare it with two Bayesian predictors (Raventós et al., 2023): one that uses the training reliability p_{train}, and one that uses the test reliability p_{test} and is therefore Bayes-optimal. Because a Bayesian predictor with reliability p predicts $p\, m^\top x$ at $n = 0$, we summarize a network's predictions by the *fitted description weight* q, the coefficient from regressing them on $m^\top x$.

For $c = 10$, networks trained at $p = 0.9$ keep description weights near 0.9, as the p_{train} predictor does, even when every test description is relevant (Table 4), and so fall short of the Bayes-optimal ESS (Figure 3b). The opposite shift, from reliable descriptions in training to unreliable ones at test time, costs little without demonstrations (Figure 3b). The two shifts differ because the Bayes-optimal error differs in size. When one description in ten is irrelevant, that error is already large, and a weight of 1 instead of 0.9 adds only about 5% to it. When every description is relevant, the error is small, and a weight of 0.9 instead of 1 roughly doubles it. With six or more demonstrations, however, the description raises the error of networks trained at $p = 1$ when they are tested at $p = 0.9$, whereas networks trained at $p = 0.99$ are not harmed (see *Reliability shift after demonstrations are observed* below).

Evaluation. We evaluate each network on $20{,}000$ fresh prompts. The networks share these evaluation prompts, so standard deviations across training seeds exclude prompt-sampling variation. At $d = 5$, we evaluate the baseline curve at every integer from 0 to 20, except for $c = 0.2$ and $c = 2$, which use counts from 0 to 15. We do not train networks at $c = 100$, the precision used in Sections 3–4: at $d = 5$, its Bayes-optimal ESS, about 106, lies far beyond the 20 demonstrations over which a network's baseline curve is measured. We compute ESS at the first crossing of this curve with the prediction error obtained using the description, as in Definition 1. Table 4 reports results at $n = 0$, including test reliabilities that differ from the training reliability.

Architecture. The Transformer has 12 layers, 8 attention heads, and width 256, matching the model size of Garg et al. (2022). Layers use pre-normalization, GELU activations, feed-forward width $1{,}024$, and no dropout. At $d = 5$, the model has 9.5 million parameters.

Tokens. The description token carries m, each demonstration token carries an input and its label, and the prediction is read from the query token. Two indicator coordinates distinguish token types, and the model uses no positional encoding. When a description is absent, its token is masked out of attention. Labels and description means are divided by τ_0.

Optimization. Following Zhu et al. (2026), the model predicts a mean and a log variance for the query label and minimizes log loss. The optimal predicted mean is the posterior mean, which also minimizes squared error, and we evaluate the predicted mean using the loss in Equation (2). We train with Adam, using the PyTorch defaults $\beta_1 = 0.9$ and $\beta_2 = 0.999$ and stability constant 10^{-8}, at a constant learning rate of 10^{-4} with no weight decay, for $40{,}000$ steps of $1{,}024$ newly sampled prompts and no curriculum. Gradients are clipped to norm 1, and training uses bf16 autocast. The demonstration count is sampled uniformly from 0 to 20, except for the $c = 0.2$ and $c = 2$ experiments, which use 0 to 15, and the $d = 10$ experiments, which use 0 to 40. We train three seeds for $c = 10$, for $d = 10$, and for the partial description; two for $c = 1$; and one each for $c = 0.2$ and $c = 2$. A $d = 5$ network takes approximately 80 minutes to train on one NVIDIA L4 GPU, and a $d = 10$ network takes approximately 2.8 hours. Training all 25 networks takes approximately 38 GPU-hours.

Without a description, every $d = 5$ network's error is within 0.21 of the Bayes-optimal error of approximately 50 at $n = 0$, and within 0.32 on average over its training demonstration counts. At $d = 10$, the gap is at most 0.50 at $n = 0$, where the Bayes-optimal error is approximately 100, and 0.41 on average over the training demonstration counts. With a description at $c = 10$, $p = 1$, and $n = 0$, network error ranges from 0.67 to 0.71, compared with 0.49 for the Bayes-optimal predictor. The networks therefore remove 98.6% of the baseline error, compared with 99.0% for the Bayes-optimal predictor.

Additional specificity levels. Four additional networks cover $c = 0.2$ and $c = 2$, each at $p = 1$ and $p = 0.9$. Their zero-shot ESS is within 0.03 of the Bayes-optimal value on the same prompts. Network and Bayes-optimal ESS are 2.64

Table 4. **Transformer ESS and fitted description weights under matched and shifted reliability.** Results use $d = 5$, $\tau_0^2 = 10$, and no demonstrations, with 20,000 test prompts per network. Rows are grouped by training reliability. *Network* reports mean ESS and standard deviation across the indicated training seeds; for a single seed, the uncertainty is the standard error over prompts. The two Bayesian predictors use p_{train} and p_{test}, respectively; the latter is Bayes-optimal under the test distribution. We show one reference value when they coincide. At $n = 0$, a network's prediction depends only on m and x, whose distribution does not change with the test reliability, so its fitted weight is the same at every p_{test}; the $p_{\text{test}} = 0.99$ rows for training reliabilities 1 and 0.9 differ only because they average one seed fewer. The values using p_{test}, and those using p_{train} where the two reliabilities coincide, use 200,000 prompts, with standard errors at most 0.1. The other values using p_{train} use the networks' 20,000 test prompts and can differ from a large-sample value by up to 0.6, which is why some exceed the Bayes-optimal value beside them. *Weight* is the fitted description weight q in $\hat{y} \approx q\, m^\top x$, averaged across the available seeds. This coefficient is unrestricted and need not be a relevance probability.

				ESS			
c	p_{train}	p_{test}	Seeds	Network	Bayes p_{train}	Bayes p_{test}	Weight q
1	1	1	2	5.85 ± 0.03	5.89	—	0.964
1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
10	1	1	3	14.26 ± 0.48	15.91	—	1.008
10	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
10	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
10	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
10	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
10	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
10	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
10	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
10	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

and 2.66 at $c = 0.2$ and $p = 1$, 2.16 and 2.15 at $c = 0.2$ and $p = 0.9$, 7.32 and 7.35 at $c = 2$ and $p = 1$, and 4.38 and 4.37 at $c = 2$ and $p = 0.9$.

Higher dimension and partial descriptions. At $d = 10$, we train three seeds with $c = 10$ and $p = 1$, and extend training and evaluation to 40 demonstrations so that the baseline curve reaches the ESS crossing. Network ESS is 17.8, compared with a Bayes-optimal value of 20.8 and the limiting expression $c + d + 1 = 21$. As at $d = 5$ with $c = 10$ and $p = 1$, the networks capture most of the description's predictive value but fall short of the Bayes-optimal ESS.

We also train networks with a partial description, as defined in Appendix A.3. The description token contains m in the first $k = 3$ of the $d = 5$ coordinates and zeros elsewhere; the remaining task coordinates follow the baseline prior. For this description, at $c = 10$ and $p = 1$, mean ESS across three seeds increases from 3.27 at $n = 0$ to 5.55 at $n = 10$. The corresponding Bayes-optimal values are 3.22 and 5.64. This increase agrees with the behavior described by Proposition 3.

ESS with demonstrations. Figure 5 evaluates each network at its training reliability. For $c = 1$, network ESS remains within 0.6 of the Bayes-optimal value at every demonstration count whose crossing lies within the evaluation grid. For $c = 10$, the largest shortfalls are 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, respectively; the largest excess is 0.3. Demonstrations can raise the ESS of an unreliable description (Section 4): for the three $c = 10$ networks trained at $p = 0.9$, ESS rises from 4.66 at $n = 0$ to between 4.77 and 4.83 at $n = 5$, and the Bayes-optimal ESS on the same prompts rises from 4.63 to 5.20.

Reliability shift after demonstrations are observed. For $c = 10$, networks trained at $p = 1$ and $p = 0.99$ both have fitted description weights near one at $n = 0$. With demonstrations, their responses to a shift in test reliability differ. At test

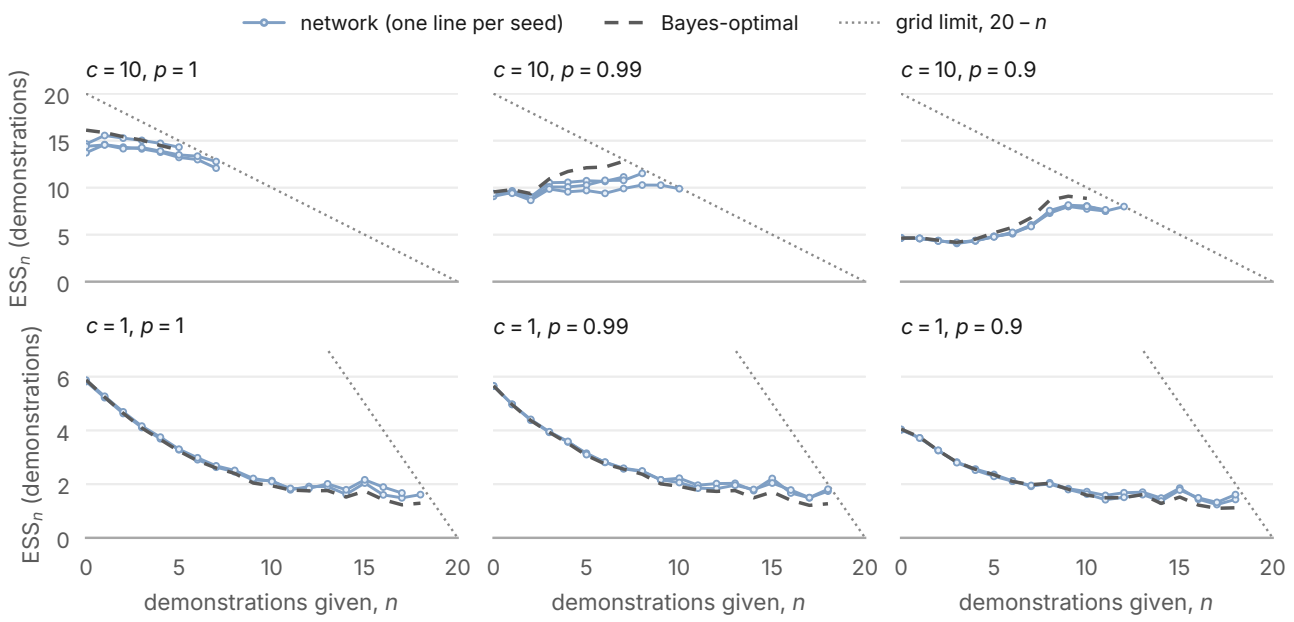


Figure 5. **Transformer ESS as a function of demonstration count.** Each line represents a training seed; the Bayes-optimal predictor is evaluated on the same prompts. Training and test reliability are matched. Results cover $c = 10$ and $c = 1$ at $d = 5$, using 20,000 prompts. A curve ends when its crossing exceeds the maximum of 20 demonstrations. The dotted line, $20 - n$, marks the largest ESS identifiable on this grid.

reliability $p = 0.9$, the description reduces error for networks trained at $p = 0.99$ at every evaluated n, as it does for the Bayes-optimal predictor under the test distribution. For networks trained at $p = 1$, the description instead increases error from six demonstrations onward in every seed. The excess error ranges from 0.77 to 1.06 at $n = 6$ and peaks between 2.74 and 2.94 at $n = 9$. A Bayesian predictor that assumes relevance ($p = 1$) also first has higher error than the baseline predictor at $n = 6$, with excess error 1.12 there and 2.96 at $n = 9$; this predictor returns the relevant component's prediction, $\mu_1^\top x$, whatever the demonstrations show. These networks behave as if every description were relevant.

C. Additional Language-Model Results and Evaluation Details

To measure behavior without reasoning more precisely and on more models, we run six open-weight models without reasoning on 2,000 tasks; we refer to these experiments as the 2,000-*task run*. The models are gpt-oss-20b, Qwen3.5 9B, and Gemma 4 31B, which also appear in Section 5.2, together with Phi-4 (14B; Abdin et al., 2024), OLMo 2 32B Instruct (Team OLMo et al., 2025), and Gemma 4 12B. The main numerical experiments in Section 5.2 use the first 200 of these tasks. The message-sorting experiment uses a separate set of 200 tasks, described in Appendix C.8. Appendices C.1–C.3 and the complete tables at the end use the 2,000-task run; the other subsections use the 200-task runs of Section 5.2.

Table 5 reports median ESS over $n \in \{1, 2, 4\}$, with uncertainty intervals. Figure 6 shows the underlying learning curves. Tables 14 and 15 report ESS at every evaluated demonstration count; Table 6 compares interpolation methods; and Table 16 reports description weights for the models with a distribution readout.

For the numerical experiments, the *Bayesian reference* combines the probabilistic statements in the prompt with an assumed baseline prior when the prompt does not specify one. It is optimal under that assumed model, but need not minimize error under the actual task distribution when the two differ. We therefore distinguish this reference from the correctly specified Bayesian predictor in the theory. The message-sorting experiment specifies a finite task distribution and uses its exact Bayesian predictor.

In the numerical task, $c = \sigma^2/\tau^2$ is the prior precision expressed in units of one demonstration. In one condition, the precision used to generate a description differs from the one stated in the prompt: descriptions generated at $c = 36$ are stated at $c = 4$ (Section 5.2); there we name both. Similarly, a stated reliability supplies a prior relevance probability to the model; it need not equal the proportion of relevant descriptions in the evaluation sample.

C.1. Prompts, Tasks, and Answer Extraction

Model inclusion. We retain a model only if its numeric response rate, defined below for the readout we use, is at least three quarters overall and, averaged over $n = 1$, 2, and 4 and both noise levels ($\sigma = 30$ and 60; see *Task generation*), at least one half without a description, since those replies determine the baseline curve used for median ESS. This criterion excludes MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct, whose replies begin with a number in 62%, 20%, and 10% of cases, respectively. These models often begin with explanations such as "To predict…" or "Given…" despite the numeric-only instruction.

The 2,000-task run uses Hugging Face Transformers at each model's native precision, with checkpoints `openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, and `google/gemma-4-31B-it`. Tasks are generated with a fixed random seed, so every model sees the same tasks. Reasoning is disabled where the chat template supports this option. gpt-oss-20b writes its reasoning in a separate channel before its answer; we start generation directly in the answer channel.

Distribution and written readouts. For gpt-oss-20b, Phi-4, and OLMo 2 32B without reasoning, the prediction is the mean of the first-token distribution, renormalized over the 900 three-digit number tokens. We call this the *distribution readout*. It is possible for these three models because their tokenizers encode each number from 100 to 999 as a single token; the other models' tokenizers split numbers into digits. Other models, and all reasoning runs, use the number in the final answer, called the *written readout*. We also evaluate written readouts for the three distribution-readout models as a robustness check (Table 5 for gpt-oss-20b, and Table 7). OLMo 2 32B assigns 41% of first-token probability to numbers without a description at $n = 1$, and 75% at $n = 2$. Its distribution readout at $n = 1$ therefore summarizes a minority of the first-token probability mass.

Table 5. **Description ESS and weights in the 2,000-task run without reasoning.** Left: median ESS over $n = 1$, 2, and 4, for relevant descriptions with $c = 1$, 4, and 36, and for the $c = 36$ description when irrelevant or accompanied by a stated 10% or 50% chance of being wrong. The stated-chance columns come from the open-alternative design of Appendix C.6, in which every description is relevant and the prompt says only that the mean is otherwise unknown; they are not the mixed design of Section 5.2. Brackets give 95% bootstrap intervals. The marker $\leq$ shows that a median is only an upper bound, because at one of its three counts a single demonstration already gives lower error than the description (Appendix C.2); such a median is rounded up so that the printed bound holds, and so can be 0.1 above the same value as rounded in Tables 14 and 15. Right: the coefficient on the prior mean at $n = 1$; the Bayesian reference weight is $c/(c+n)$. All results use $\sigma = 60$. "Written" denotes the written readout.

	ESS, median over $n = 1, 2, 4$						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	10% wrong	50% wrong	$c = 1$	$c = 4$	$c = 36$
Reference	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

Prompt. The following text is supplied as the user turn in the model's chat template:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.
548
431
Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." To state unreliability, a third sentence is added. In the mixed design of Section 5.2 (Appendix C.6), it reads "There is a 10% chance that this statement is wrong, in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999." In the open-alternative design

of Appendix C.6, its final clause is "in which case the mean is unknown." With no demonstrations, the final question is "Predict the first number."

Task generation. A relevant description's prior mean is sampled conditional on the task so that the joint distribution satisfies $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$. Demonstrations are rounded to integers. Language models predict short unsigned numbers best (Gruver et al., 2023; Requeima et al., 2024), so we keep every number to three digits: a task is redrawn if a displayed number or prior mean falls outside 100 to 999, which occurs for less than 1% of tasks. The 2,000-task run also includes $\sigma = 30$ conditions. Both standard deviations stated in the prompt are then halved, preserving $c \in \{1, 4, 36\}$. The qualitative pattern is unchanged (Table 14).

Numeric response rates. For the distribution readout, the numeric response rate is the probability mass assigned to number tokens at the first position. For the written readout, it is the fraction of replies containing a three-digit number. Tables 7, 14, 15, and 16 report this quantity as "number". Tables 7 and 16 italicize cells whose rate is below three quarters.

Written-readout implementation and checks. The written readout extracts the number in the last `\boxed{}` expression in the final answer, if present. Otherwise, it takes the last number with three digits before an optional decimal part, retaining the decimals. A digit group after a thousands separator, as in 14,945, does not count as a qualifying number. In the 2,000-task run, decoding is greedy with a limit of eight new tokens, so the reply is the number itself; for these replies, the readout takes the first qualifying number and keeps only its integer part. Replies without a qualifying number are excluded from both error curves and weight regressions; the reported response rates identify the extent of this exclusion.

For gpt-oss-20b, written-readout weights differ from distribution-readout weights by at most 0.21 across all conditions and demonstration counts. For $c = 1$, 4, and 36, written-readout median ESS is 0.2, 2.0, and 3.7, compared with 0.9, 2.0, and 2.8 from the distribution readout. A single greedy response is a noisier summary than the distribution mean. Phi-4 produces a number in only 58% of written replies across all conditions, so Table 5 reports only its distribution readout. In the wording experiment of Appendix C.3, however, its numeric response rate exceeds three quarters for the original wording, so Table 7 includes its written replies. OLMo 2 32B produces a number in only 5%, 36%, and 48% of written replies without a description at $n = 1$, 2, and 4, respectively, with $\sigma = 60$; we therefore do not report its written-readout ESS. Its written-readout weights for the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, compared with distribution-readout weights of 0.72, 0.70, and 0.70.

C.2. ESS Estimation and the Bayesian Reference

Measurement. We apply Definition 1 to squared error against the hidden mean. In the 2,000-task run, the baseline curve is evaluated at $n = 1, 2, 4, \ldots, 64$. Because this grid doubles n, we interpolate squared error linearly in $\log n$, which is less biased here than interpolation in n (see *Sensitivity to interpolation* below). The ESS at n is the crossing n^*, the first interpolated demonstration count whose baseline error matches the error with the description, minus n. For a decreasing baseline curve, a negative ESS indicates higher error with the description than without it at the same demonstration count. For the Bayes-optimal predictor under the correctly specified model, conditioning on a description cannot increase expected error, so its ESS is nonnegative.

If the error with the description lies below the entire baseline curve, which ends at $t_{\max} = 64$, we report $\mathrm{ESS}_n > t_{\max} - n$. Because the curve starts at one demonstration, we use the marker $\mathrm{ESS}_n \leq 1 - n$ when one demonstration alone already gives lower error. These markers record censoring at the evaluation boundaries; they do not define an extrapolated matching count or an exact ESS when the crossing does not exist. Median ESS uses $n = 1$, 2, and 4. We exclude $n = 0$ because models rarely return a number without a description or demonstrations. At large n, ESS is strongly affected by slow baseline learning and excessive weight on the $c = 1$ and $c = 4$ descriptions. At $\sigma = 60$ and $n = 64$, model root-mean-square error without a description ranges from 9 to 20, compared with 7.5 for the reference. A learner with a flatter baseline curve can assign a larger ESS to the same error reduction. Description-weight estimates complement ESS because they do not use that curve. We obtain them by regressing predictions on m, $\bar{y}$, and an intercept: the coefficient on m is the *fitted description weight*, and the reference weight is $c/(c + n)$, which ranges from 0.2 to 0.97 over the conditions we evaluate.

Bootstrap replicates resample tasks jointly for the curves with and without the description and recompute the complete ESS statistic. The intervals therefore measure variation across tasks; they do not include interpolation bias, quantified below. We use 500 replicates with a fixed seed and the same resampled tasks for every model.

The reference's baseline prior and its exact ESS. Because every input is the same, the $d + 1$ term of Section 3, which arises from random inputs, does not arise here: in the idealized continuous Gaussian model with a flat baseline prior, a correctly specified description has exact ESS c, its conventional prior ESS (Neuenschwander et al., 2020). The Bayesian reference takes "unknown" to mean uniform over 100 to 999. Without a description, its prediction is therefore approximately the sample mean, with negligible boundary truncation for the evaluated prompts. For a relevant Gaussian description with correctly stated precision, the idealized continuous model with a flat baseline prior gives exact ESS c. Integer rounding, finite-range truncation, and the task-rejection rule are not part of that exact identity; we compute the reported reference values on the same generated tasks as the models. When a chance of being wrong is stated, the reference uses it as the prior probability of irrelevance and computes the posterior relevance probability from the demonstrations. Otherwise, it treats the description as relevant.

Applying the empirical ESS procedure to the reference on the 2,000-task run gives 1.3, 4.4, and 36 for $c = 1$, 4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$. Interpolation in $\log n$ induces an upward bias, while finite-task sampling can shift the estimate in either direction. We use the same estimator for the reference and the models. On the 200 tasks of Section 5.2, the grid additionally includes $n = 40$ and 48 near the $c = 36$ crossing, and the reference values are 1.0, 3.5, and 34.6. In the open-alternative design of Appendix C.6, in which every description is relevant and the prompt says only that the mean is otherwise unknown, the reference's median ESS varies more across task samples: on 20,000 tasks it is 7.3 at a stated 50% and 23.4 at 10%, compared with 5.8 and 22 on all 2,000 tasks and 5.7 and 27.5 on the first 200. Each model is compared with the reference on the same tasks.

The actual task prior, $\mathcal{N}(500, 100^2)$, is not stated in the prompt. A reference using that prior would have baseline precision equivalent to 0.36 demonstrations at $\sigma = 60$ and 0.09 at $\sigma = 30$. At $\sigma = 60$, the corresponding idealized continuous Gaussian model would give a relevant description exact ESS $c - 0.36$, namely 0.64, 3.64, and 35.64. This is the scalar analogue of $c - c_0$ in Section 3. We use the Bayesian reference because the models are not given the actual task prior. This distinction matters under misspecification: unlike the correctly specified Bayesian predictor in Sections 3–4, a reference that incorrectly assumes an irrelevant description to be relevant can have negative ESS, indicating increased prediction error (Reimherr et al., 2021).

Sensitivity to interpolation. The grid of the 2,000-task run doubles n, so interpolation can affect ESS. On the reference's idealized errors, σ^2/n without a description and $\sigma^2/(c + n)$ with one, interpolation in $\log n$ overestimates ESS by 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36. Linear interpolation in n gives overestimates of 0.3, 0.6, and 4.1.

For the models without reasoning, every crossing is unique, because every baseline curve decreases over the counts where crossings occur. Among the twelve baseline curves across the two noise levels, the only increase is Phi-4's at $\sigma = 60$, whose root-mean-square error rises slightly from 20.3 at $n = 32$ to 20.4 at $n = 64$. Table 6 compares four estimators: interpolation in $\log n$; linear interpolation in n; a least-squares non-increasing fit using the pool-adjacent-violators algorithm; and a relative-least-squares fit of $A/(n + n_0) + B$ to the baseline errors. This last form is a Bayesian learning curve plus an error floor and recovers exact ESS when fit to the reference's idealized errors.

Across these choices, no model's median ESS changes by more than 1.4 demonstrations. For the $c = 36$ description, no model exceeds 8.1, while the reference ranges from 35 to 38. Differences become larger at higher demonstration counts, where a flat learning curve turns a small error difference into a large change in ESS. For example, gpt-oss-20b at $n = 8$ and $c = 36$ has ESS 17 under the primary log-n estimator (Table 14), but 9.2 under the fitted curve.

C.3. Irrelevant Descriptions and Wording

Apart from the stated-reliability experiments of Section 5.2, the experiments in the main text keep every description relevant, and all of them state its precision in one wording. This section relaxes each of these choices in the 2,000-task run.

Irrelevant descriptions. An irrelevant description states a prior mean drawn independently of the task, from the same distribution as relevant prior means, with the same stated precision. At $c = 36$, these descriptions have negative ESS for every model and for the Bayesian reference, which treats them as relevant. The models assign less weight to the description and consequently incur lower error in this condition: at $n = 4$, their root-mean-square errors range from 35 to 64, compared with 124 for the reference.

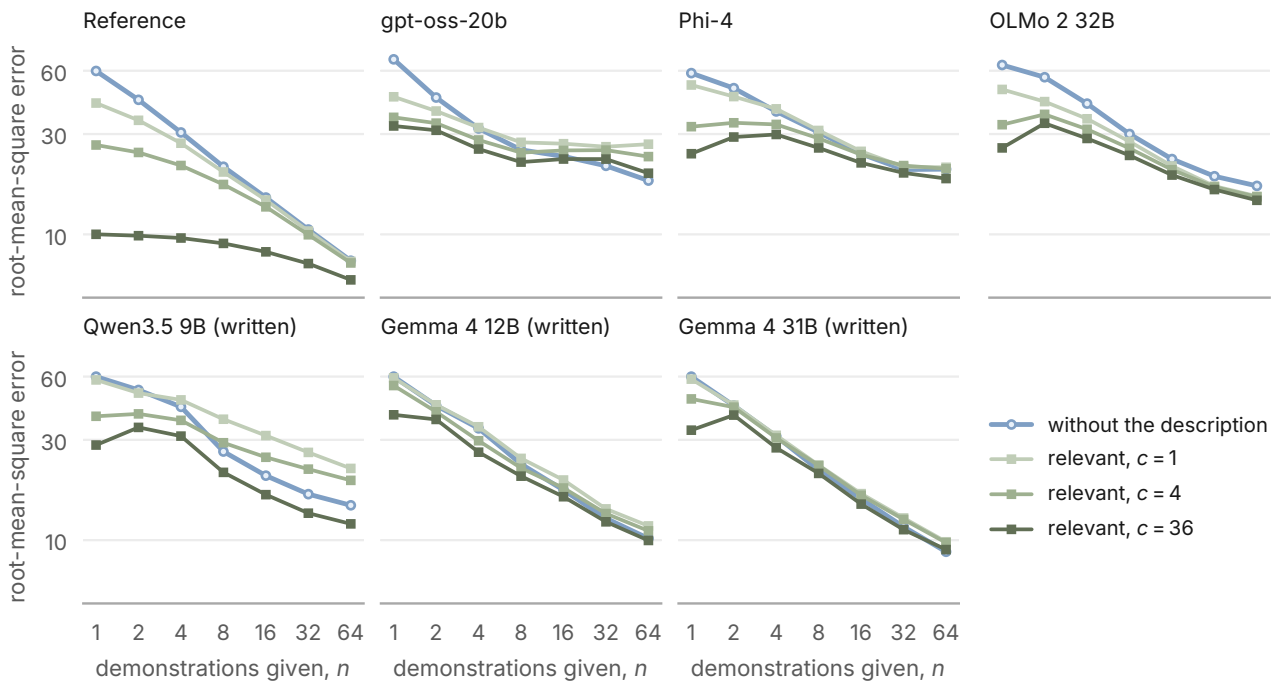


Figure 6. **Learning curves used to estimate language-model ESS.** Root-mean-square error versus demonstration count for the reference and six models in the 2,000-task run without reasoning, at $\sigma = 60$. Blue curves omit the description; green curves use relevant descriptions, with darker shades indicating higher c. Replies without a number are excluded. ESS is the horizontal distance from the point with the description at n to the baseline curve at the same error. For Phi-4, OLMo 2 32B, Qwen3.5 9B, and Gemma 4 31B, error with $c = 36$ increases between one and two demonstrations because the models shift toward the observations more than the reference does; Table 16 shows this for the first two.

Wording without reasoning. Descriptions in practice express uncertainty in many ways. To test whether the results depend on the original wording, we repeat the three relevant-description conditions and the no-description baseline at $n = 1$, 2, and 4 using three alternative description sentences: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." Table 7 uses the tasks of the 2,000-task run and written readouts.

Wording substantially changes the weight assigned to the description. For Gemma 4 12B with $c = 1$ and $n = 1$, this weight is 0.34 under the original wording and 1.00 under each alternative. For gpt-oss-20b, it ranges from 0.17 to 0.67. Nevertheless, none of the tested wordings yields the reference's sensitivity to prior precision: across models and wordings, the weight difference between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$, compared with reference differences of 0.47 and 0.70. The informal descriptions do not uniquely specify a probability distribution, so this comparison changes how explicitly the prompt specifies the prior, not only its wording.

C.4. Reasoning Settings and Inference Details

Comparisons between modes. The labels "with reasoning" and "without reasoning" refer to generation settings. When we say that a reply contains reasoning, we mean that the provider recorded reasoning output for it; with reasoning enabled, some replies have none. Paired runs use the same model identifier and prompts, but reasoning settings, serving software, readout, and retry policy vary across the runs. These comparisons concern the evaluated configurations and do not identify a unique internal mechanism.

Shared evaluation prompts. Most configurations in Figure 4 are evaluated on the same 7,800 prompts from the first 200 tasks. These include no-description prompts at $n = 1, 2, 4, \ldots, 64$, with additional points at $n = 40$ and 48 near the $c = 36$ crossing, and every description condition at $n = 1$, 2, and 4. The gpt-oss-20b and Olmo 3 7B Think (Team Olmo, 2025) runs omit the two additional baseline points and contain 7,400 prompts. Results without reasoning for the open-weight models are taken from the 2,000-task run restricted to these tasks. Paired mode comparisons use the same prompts, although serving environments differ as detailed below. At least 99.7% of replies in each run contain a number.

Table 6. **Sensitivity of median ESS to baseline-curve estimation.** Median ESS over $n = 1$, 2, and 4 for relevant descriptions, using the 2,000-task run without reasoning at $\sigma = 60$. "Log n" is the primary estimator; "linear" interpolates in n; "monotone" applies a non-increasing fit; and "fit" uses $A/(n + n_0) + B$. The reference's idealized ESS values are 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Reference	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

Table 7. **Description weights under four wordings without reasoning.** Weights are reported for the three relevant descriptions at $n = 1$ and 4, using written readouts on the tasks of the 2,000-task run. "Number" denotes the numeric response rate averaged over demonstration counts; a weight is italicized when the response rate at that count is below three quarters. The reference weights are 0.50, 0.80, and 0.97 at $n = 1$, and 0.20, 0.50, and 0.90 at $n = 4$. A fitted weight above 1 means that, holding the sample mean fixed in the regression, a unit change in the prior mean corresponds to more than a unit change in the prediction. Wording changes overall reliance on the description, but the observed change with stated precision remains smaller than the reference change.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (ours)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (ours)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (ours)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (ours)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (ours)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (ours)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

Open-weight reasoning models. With reasoning, gpt-oss-20b runs in vLLM using greedy decoding, low reasoning effort, and at most 4,096 new tokens. It reaches the final channel in 99.9% of replies and uses 322 tokens on average. Qwen3.5 9B runs through OpenRouter at temperature 0, with reasoning enabled and a limit of 24,576 new tokens. We pin requests to DeepInfra, which serves the model in bf16, because quantization varies across providers. Of its replies, 2,305 repeat themselves until they reach the token limit. Each is rerun once with the model card's recommended reasoning sampling settings: temperature 1.0, top-p 0.95, top-k 20, and presence penalty 1.5. Eleven remain unfinished and one request fails.

Gemma 4 31B (`gemma-4-31b-it`) runs through the Gemini API at temperature 0. Reasoning is enabled by default, and

the model uses 2,613 reasoning tokens on average, with the same 24,576-token limit. Of its replies, 435 reach the limit; all finish after one rerun with the model card's recommended sampling settings: temperature 1.0, top-p 0.95, and top-k 64.

We also run Olmo 3 7B Think (`allenai/Olmo-3-7B-Think`), a small open-weight model that always reasons and so has no run without reasoning. It runs in vLLM with greedy decoding and at most 32,768 new tokens. We rerun the 169 replies that reach this limit once with temperature 0.6 and top-p 0.95; 17 remain unfinished. The model generates 9,215 tokens on average and writes a worked solution before most answers. Its long greedy outputs depend on the hardware: an earlier run of the same conditions on a different GPU gave the same final number on only half the prompts. We report a single run and treat it as one sample from this variation. It averages the demonstrations about as well as the reference but places more weight on the description at $c = 1$ and 4. At $c = 36$, its replies split between the prior mean alone and other prediction rules, as detailed below.

Models served through command-line agents. GPT-6 Sol (`gpt-6-sol`, the model identifier served by Codex during the runs) runs through the Codex command-line agent in a read-only sandbox, with an empty configuration directory so that no user instructions or settings load, and with every tool that can run code disabled. We evaluate the model with reasoning disabled (no reasoning tokens) and enabled at low effort, at which, in a pilot, its answers were already within 1 of the reference's predictions.

Claude Opus 4.6 (`claude-opus-4-6`) and Haiku 5.5 (`claude-haiku-5-5`) run through the Claude Code command line, with one prompt per call, no tools, no project or user instructions, and the system prompt "You are a helpful assistant." Opus 4.6 uses high effort in both evaluated modes, with reasoning either enabled or disabled, so that the two runs differ only in reasoning; the disabled mode emits no reasoning tokens. With reasoning enabled, at least 99% of replies in each description condition include reasoning output, with an average of 599 reasoning tokens across all replies. Without a description, only 1%, 24%, and 79% of replies include reasoning output at $n = 1$, 2, and 4, respectively, so its baseline curve at small n mostly reflects replies without reasoning; using the reference's baseline curve instead changes its ESS by at most 1 (see below). Haiku 5.5 is evaluated mainly at medium effort, because at low effort it often answers without reasoning; we also report low effort for comparison. At low effort and $n = 1$, it often repeats the single demonstration value without reasoning output, so its description weight at $n = 1$ is low even in conditions whose ESS is close to the reference value. At medium effort and $c = 36$, its ESS is 3.4 at $n = 1$, 35 at $n = 2$, matching the reference, and 28 at $n = 4$, against the reference's 33; its median ESS is therefore 28, against 34.6, though its bootstrap interval in Figure 4b includes the reference's value. At $n = 1$, 28% of its replies contain no recorded reasoning output, against 19% at $c = 1$ and 20% at $c = 4$. Both Claude models use the command line's default sampling settings, which the command line does not report, and we collect one reply per prompt. Without reasoning, Opus 4.6 still writes out intermediate steps in 64 of its 7,800 replies; excluding these replies changes no median ESS by more than 0.1.

Run dates. Because hosted models can change over time, we record when each was evaluated. The 2,000-task run took place on 2–3 October 2026, and its conditions in which the stated precision differs from the generating one, and gpt-oss-20b's reasoning run, took place on 20 October. Haiku 5.5 was evaluated on 20 October for the runs of Section 5.2 and 4 November for wording, the stated baseline prior, and the message-sorting experiment. Opus 4.6, Qwen3.5 9B, Gemma 4 31B, and Olmo 3 7B Think were evaluated on 3–4 November. GPT-6 Sol was evaluated on 4 November. The irrelevant-description runs of the mixed design took place on 9–10 November, when Opus 4.6's runs with the baseline prior stated were also completed. The GPT-6 Sol logs do not record a model version beyond the identifier above.

Without reasoning. Counted in demonstrations, the shortfall from the reference is much smaller at $c = 4$ and $c = 1$ than at $c = 36$, because the reference's own ESS is small there, but most models still reach less than three fifths of the reference's ESS (Table 8). Most models also respond too little to the stated precision itself. When the prompt states $c = 4$ for descriptions generated at $c = 36$, the reference's ESS falls from 34.6 to 13.7 (Table 8). It stays above its value for descriptions generated at $c = 4$, because the prior mean is as accurate as before. GPT-6 Sol's ESS falls by a larger fraction than the reference's, and that of Opus 4.6 and Gemma 4 31B by about half, somewhat less than the reference's. The ESS of Qwen3.5 9B and gpt-oss-20b falls by a quarter or less, against three fifths for the reference, so these two models respond little to the stated precision.

ESS when a model's baseline curve stops falling. With reasoning, gpt-oss-20b and Qwen3.5 9B nearly match the reference description weight: at $n = 4$ and $c = 36$, their fitted weights are 0.89 and 0.90, compared with 0.90 for the reference. Their errors without descriptions stop falling at around 32 demonstrations, so at two or more of the three counts

Table 8. **Description ESS across reasoning settings.** Results use the first 200 tasks at $\sigma = 60$. Median ESS uses each run's own baseline curve; parentheses report the value computed using the reference baseline where it differs. "Stated irrelevance" gives the stated probability that a description generated at $c = 36$ is irrelevant, in the open-alternative design of Appendix C.6: every description is in fact relevant, and the prompt says only that the mean is otherwise unknown. These columns therefore differ from the mixed design of Section 5.2, whose results are in Table 10. The last column fixes the generating precision at 36 and states it as 4. Each run names the mode and, in parentheses, the provider's effort setting. An upper bound, $\leq$, indicates that one demonstration alone gives lower error; > 60 means that the error with the description lies below the model's entire baseline curve, through $n = 64$, so no crossing exists on the grid. Reference values use the same tasks and estimator. The open-weight rows without reasoning take these 200 tasks from the 2,000-task run, whose results on all its tasks are in Table 5. The runs also include three further conditions in which the stated precision differs from the generating one, not shown.

					Stated irrelevance		
Run	$c = 1$	$c = 4$	$c = 36$	Irrelevant	10%	50%	Stated $c = 4$ Generated $c = 36$
Reference	1.0	3.5	34.6	≤ -1.0	27.5	5.7	13.7
GPT-6 Sol, without reasoning	0.9	1.9	10	≤ -1.0	3.3	0.9	2.4
GPT-6 Sol, with reasoning (low)	0.9	3.3	35 (34)	≤ -1.0	30 (29)	4.2	11
Opus 4.6, without reasoning	0.9	2.0	5.8 (5.9)	≤ -1.0	6.1 (6.0)	3.8	3.0
Opus 4.6, with reasoning (high)	1.0	3.5	35	≤ -1.0	29	3.8	13
Gemma 4 31B, without reasoning	0.0 (-0.2)	0.5 (0.2)	2.0 (1.6)	≤ -0.3	1.0 (0.7)	0.8 (0.4)	1.0 (0.6)
Gemma 4 31B, with reasoning	1.0	3.5	35	≤ -1.0	21	4.0	14
Qwen3.5 9B, without reasoning	-0.2 (-0.8)	2.7 (0.4)	5.5 (1.1)	≤ -1.0	1.6 (-0.7)	2.3 (-0.6)	4.1 (1.1)
Qwen3.5 9B, with reasoning	0.9	3.5	> 60 (34)	≤ -1.0	29 (27)	3.9	15 (14)
gpt-oss-20b, without reasoning	0.5 (0.3)	1.8 (1.3)	2.7 (1.7)	≤ -0.5	1.8 (1.5)	1.7 (1.5)	2.5 (1.6)
gpt-oss-20b, with reasoning (low)	0.3	1.0	> 60 (24)	≤ -1.0	1.3 (1.2)	8.8	8.6 (8.4)
Haiku 5.5, with reasoning (medium)	1.1	3.3 (3.5)	28	≤ -1.0	33	3.8 (3.9)	8.5 (8.8)
Haiku 5.5, with reasoning (low)	0.8 (0.7)	3.2 (3.3)	24	≤ -1.0	27 (26)	5.5 (5.6)	8.6 (8.4)
Olmo 3 7B Think, with reasoning	-0.5	0.0	6.2 (6.0)	≤ -1.0	5.9 (5.6)	5.5 (5.3)	15 (13)

we evaluate, their error with a $c = 36$ description lies below their entire baseline curve, and their median ESS on their own curves is not defined (> 60 in Table 8). gpt-oss-20b makes errors when averaging long sequences: its no-description root-mean-square error is 19.0 at $n = 32$ and 24.7 at $n = 64$, compared with reference errors 10.0 and 7.4. At these counts, 37 and 62 replies, respectively, differ from the sample mean by more than 20. Qwen3.5 9B levels off near 11 from $n = 32$ onward (11.2 at $n = 64$), with up to 11 replies per cell that far from the sample mean.

For these models, the error with a description sometimes lies below the whole baseline curve, so the evaluated grid contains no crossing. Table 8 therefore also reports, in parentheses, a comparison using the reference's baseline. Specifically, it finds the reference demonstration count t_{ref} that matches the model's error with the description and reports $t_{\text{ref}} - n$. This comparison differs from learner-specific ESS because it uses another predictor's baseline curve. gpt-oss-20b also makes arithmetic errors with descriptions: about 2% of its replies are more than 40 from the reference prediction, from errors such as exchanging the description's and the demonstrations' precisions. These errors reduce its ESS; for example, its median ESS is 1.3 when the prompt states a 10% chance of irrelevance in the open-alternative design of Table 8. At a stated 50%, by contrast, it largely ignores the stated chance (Table 11), and because every description in those runs is relevant, its ESS of 8.8 exceeds the reference's 5.7.

Opus 4.6 and GPT-6 Sol do not exhibit these arithmetic failures. Opus 4.6 has root-mean-square error 7.4 at $n = 64$, matching the reference on these tasks. For these two models, changing between the model and reference baselines changes ESS by at most 1. Table 9 reports each run's absolute error with a description.

C.5. Prediction Rules in Individual Replies

Per-response description weights. At $n = 4$, we compute the per-response description weight $(\hat{w} - \bar{y})/(m - \bar{y})$. This quantity describes one response and differs from the fitted coefficient obtained by regressing predictions across tasks. We use the tasks with $|m - \bar{y}| \geq 30$, where the weight is well determined: 61 tasks at $c = 36$, whose more accurate prior means lie closer to the sample mean, and 124 at $c = 1$. We exclude replies without a number. A reply is classified as consistent with

Table 9. **Absolute error with a description.** Root-mean-square error against the hidden mean, for descriptions generated and stated at $c = 4$ and $c = 36$, on the first 200 tasks at $\sigma = 60$. Rows are the runs of Table 8, whose ESS compares each run with its own baseline curve; this table reports the error itself. Replies without a number are excluded.

	$c = 4$			$c = 36$		
Run	$n = 1$	2	4	1	2	4
Reference	26.7	24.0	19.9	9.5	9.3	9.3
GPT-6 Sol, without reasoning	32.1	28.6	24.2	15.2	16.3	18.4
GPT-6 Sol, with reasoning (low)	26.7	26.7	20.1	9.5	9.4	9.6
Opus 4.6, without reasoning	29.4	28.1	23.7	13.6	18.9	18.1
Opus 4.6, with reasoning (high)	26.7	24.0	19.8	9.5	9.3	9.3
Gemma 4 31B, without reasoning	43.4	41.6	27.6	32.6	39.1	24.0
Gemma 4 31B, with reasoning	26.7	24.0	19.9	9.5	9.3	9.3
Qwen3.5 9B, without reasoning	38.0	38.7	34.5	28.5	33.6	26.0
Qwen3.5 9B, with reasoning	26.7	24.0	19.9	9.6	9.3	9.3
gpt-oss-20b, without reasoning	34.5	32.6	26.5	31.5	30.5	23.7
gpt-oss-20b, with reasoning (low)	28.0	34.5	52.4	11.9	32.2	11.0
Haiku 5.5, with reasoning (medium)	33.7	24.0	19.8	27.0	9.3	10.1
Haiku 5.5, with reasoning (low)	46.1	24.0	20.2	41.9	9.7	11.1
Olmo 3 7B Think, with reasoning	28.7	42.1	32.1	20.8	16.5	28.8

a rule when its per-response weight is within 0.05 of that rule's weight. The candidate weights are the Bayesian reference (0.90 for $c = 36$ and 0.20 for $c = 1$); the reference with the description's and the demonstrations' precisions exchanged, $n/(c + n)$ (0.10 and 0.80); the description counted as one additional observation, $1/(n + 1)$; the description alone, with weight 1; and the demonstrations alone, with weight 0. The one-additional-observation rule coincides with the reference at $c = 1$.

With reasoning, Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Haiku 5.5 match the reference rule on 89 to 100% of eligible replies. Without reasoning, 77% of Opus 4.6 replies at $c = 36$ match none of these candidate rules, as do more than half of GPT-6 Sol replies. Enabling reasoning does not ensure agreement with the reference in every model. Olmo 3 7B Think reaches a median ESS of only 6.2 at $c = 36$, although it averages demonstrations well: without a description, its root-mean-square error falls from 56 at $n = 1$ to 10.8 at $n = 32$, compared with 10.0 for the reference. Its shortfall lies in combining the description with the demonstrations. At $n = 1$, it repeats the prior mean in 57 to 80% of replies, more often at higher stated precision, and at $n = 4$, about half of its eligible replies return the prior mean alone and a third match the reference.

C.6. Stated Reliability

We test stated reliability in two designs. In both, the prompt adds a sentence, the *warning*, that states a chance that the description is wrong. In the *mixed design* of Section 5.2, descriptions are irrelevant with the stated probability and the warning states what holds in that case. In the *open-alternative design*, the warning leaves the irrelevant case open, saying only that the mean is then unknown; in the results of Tables 5 and 8, every description in this design is relevant; we report it second, because it shows what models do when the prompt does not determine how to update.

Mixed design. The prompt adds the sentence "There is a 10% chance that this statement is wrong, in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999," or the same sentence with 50%. For cost reasons, we run this design on the three models that our subscriptions cover, GPT-6 Sol, Opus 4.6, and Haiku 5.5. Each of the 200 tasks is evaluated with its relevant $c = 36$ description and with an irrelevant one, whose mean is drawn independently of the task from the same marginal distribution. With stated chance $1 - p$, the error with the description is p times the mean squared error with the relevant description plus $1 - p$ times that with the irrelevant one, so that the stated chance is the true one. The stated alternative is the reference's baseline prior, not the distribution that generates the tasks; the prompt states it because without it the posterior relevance probability is not determined. ESS is then computed as in Section 5.2, against each model's own baseline curve, with intervals that resample tasks. *Fixed-weight averaging* conditions each component on the demonstrations and averages the two posterior means with the prior weights p and $1 - p$; a predictor that ignores the warning treats every description as relevant.

Table 10 reports the results. At a 10% chance, the reference's median ESS is 5.0, against 34.6 without the warning, and

the two simpler predictors have ESS 0.3 and −0.1: the description's remaining value depends on recognizing, from the demonstrations, the tasks on which it is irrelevant. At a 50% chance, the reference's ESS is 0.9, and both simpler predictors do worse with the description than without it. GPT-6 Sol with reasoning is within 0.2 of the reference at both chances, and about nine in ten of its replies are within 2 of the reference's prediction with either kind of description. Opus 4.6 with reasoning is within 0.1 of the reference at both chances, with 96 to 100% of its replies within 2 of the reference's prediction. Haiku 5.5 falls short, at a 10% chance mainly with irrelevant descriptions and at 50% with both kinds. Without reasoning, GPT-6 Sol and Opus 4.6 gain little or nothing from the description.

Table 10. **Unreliable descriptions with the stated chance of irrelevance.** Median ESS in the mixed design, with 95% bootstrap intervals over tasks, and the share of replies within 2 of the reference's prediction when the description is relevant (rel.) and irrelevant (irr.), pooled over $n = 1$, 2, and 4. A value of ≤ -1.0 is censored: at $n = 2$ the description does worse than one demonstration alone.

	10% irrelevant			50% irrelevant		
	ESS	rel.	irr.	ESS	rel.	irr.
Reference	5.0 [3.9, 6.2]			0.9 [0.5, 1.2]		
Fixed weights	0.3 [−0.2, 1.0]			−0.7 [−1.0, −0.4]		
Ignores the warning	−0.1 [−0.6, 0.6]			≤ -1.0		
GPT-6 Sol, low effort	4.8 [3.8, 6.3]	89%	92%	0.8 [0.5, 1.2]	90%	92%
Opus 4.6, high effort	5.0 [3.9, 6.9]	100%	96%	0.9 [0.5, 1.3]	98%	96%
Haiku 5.5, medium effort	3.2 [2.4, 4.1]	85%	60%	0.6 [0.3, 1.0]	58%	52%
GPT-6 Sol, without reasoning	0.9 [0.6, 1.2]	19%	12%	−0.1 [−0.2, 0.1]	10%	12%
Opus 4.6, without reasoning	1.0 [0.5, 1.6]	15%	11%	0.0 [0.0, 0.0]	14%	13%

When the prompt leaves the irrelevant case open. In the open-alternative design, the warning gave a 10% or 50% chance that the description is "wrong," in which case "the mean is unknown," and the descriptions themselves remained relevant. A stated 50% chance of irrelevance then reduces reference ESS from 34.6 to 5.7 under the uniform baseline prior (Table 8). Without reasoning, the ESS of Opus 4.6, gpt-oss-20b, Qwen3.5 9B, and Gemma 4 31B falls by about a third to three fifths, against 84% for the reference, and GPT-6 Sol's falls from 10.3 to 0.9.

Fixed-weight averaging. The Bayesian reference averages its two component predictions with the posterior relevance probability π_n, which the demonstrations move away from the stated p. For $c = 36$, we compare it with fixed-weight averaging, defined above, which keeps the prior weights p and $1 - p$. Treating the uniform baseline prior as flat, fixed-weight averaging gives weight $p\,36/(36 + n)$ to m. Table 11 reports fitted description weights under the open-alternative design's warning, which leaves the baseline prior unspecified. At the stated 50%, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B with reasoning are close to fixed-weight averaging. Opus 4.6 starts near the reference at $n = 1$ but falls below fixed-weight averaging by $n = 4$, and Haiku 5.5 at medium effort lies between the two at $n = 1$ and 2. Haiku 5.5 at low effort comes closest to the reference, although its replies often contain no reasoning output at $n = 1$. The individual replies show the same pattern. Pooling $n = 1$, 2, and 4 at the stated 50%, the share of replies within 2 of the fixed-weight prediction is 100% for Gemma 4 31B, 97% for Qwen3.5 9B, 89% for GPT-6 Sol, 80% for Opus 4.6, 58% for Haiku 5.5 at medium effort, and 21% for gpt-oss-20b. The share within 2 of the reference prediction is at most 24% for every model. Regressing the replies' normalized description weights (per-response weights divided by $c/(c + n)$, defined below) on the reference's π_n gives slopes of 0.01 for GPT-6 Sol, 0.02 for Gemma 4 31B, 0.06 for Qwen3.5 9B, 0.22 for Haiku 5.5, 0.30 for Opus 4.6, and 0.00 for gpt-oss-20b, where updating as the reference does gives 1.

Stating the baseline prior. Computing π_n requires the marginal likelihood of the demonstrations under each component, and hence the baseline prior under irrelevance. The open-alternative design's warning says only that the mean is otherwise "unknown," so the prompt does not determine a unique posterior relevance probability: a model must supply the prior from its pretrained knowledge, and the reference must assume one. The assumption matters: with its baseline curve held fixed, the reference's median ESS at the stated 50% is 5.7 under our uniform prior over 100 to 999, 4.6 under the generating distribution $\mathcal{N}(500, 100^2)$, and 13.4 under a uniform prior over 0 to 2000. To remove this ambiguity, we rerun the stated-reliability condition on the same 200 tasks, at $c = 36$, stated irrelevance probabilities of 10% and 50%, and $n = 1$, 2, and 4. The prompt now states the reference's uniform baseline prior over 100 to 999. We use GPT-6 Sol at low effort and Haiku 5.5 at medium effort, with 1,200 prompts each; these runs also supply the relevant-description half of the mixed design above. Opus 4.6, at high effort with reasoning and without it, answers the same prompts, which supply the relevant

Table 11. **Description weights under stated unreliability.** Fitted coefficients on the $c = 36$ prior mean for the first 200 tasks, with stated irrelevance probability $1 - p$. All model rows use reasoning. "Fixed weights" uses weight $p\,36/(36 + n)$; the reference updates the relevance probability using a uniform baseline prior over 100 to 999. Bootstrap standard errors are at most 0.03 at $n = 4$ in the stated-50% condition and at most 0.06 overall, except for gpt-oss-20b, whose maximum is 0.09. At the stated 50%, several reasoning configurations closely track fixed-weight averaging.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol	0.49	0.46	0.40	0.90	0.86	0.71
Opus 4.6	0.56	0.43	0.40	0.93	0.86	0.77
Gemma 4 31B	0.49	0.46	0.45	0.89	0.68	0.76
Qwen3.5 9B	0.49	0.46	0.43	0.88	0.84	0.80
Haiku 5.5, medium	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, low	0.49	0.62	0.58	0.42	0.94	0.88
gpt-oss-20b	0.89	0.87	0.82	0.95	0.87	0.90

half of its mixed-design results. Table 12 compares their fitted description weights with those from the open-alternative design.

At the stated 50%, the reference and fixed-weight predictions differ by at least 5 on 123 to 155 tasks per cell. On these tasks, when the baseline prior is stated, the share of replies closer to the reference than to the fixed-weight prediction rises from 0–5% to 92–97% for GPT-6 Sol, and from 21–41% to 87–95% for Haiku 5.5. At the stated 10%, at most 7 tasks per cell separate the two rules by this amount, so the comparison is uninformative there. Pooling the three values of n at the stated 50%, the share of replies within 2 of the reference prediction rises from 10% to 90% for GPT-6 Sol and from 23% to 58% for Haiku 5.5.

A constant increase in the description weight could also make replies closer to the reference, because its posterior relevance probability exceeds the stated 50% on most tasks. To test whether the weights vary with the evidence, we use each reply's *normalized description weight*: its per-response weight divided by $c/(c+n)$, restricting to tasks with $|m - \bar{y}| \geq 30$. For the scalar reference, this equals its posterior relevance probability π_n, up to the negligible boundary effects discussed above. For model predictions, the ratio can lie outside $[0, 1]$ and does not identify an internal probability. Here π_n ranges from 0.08 to 0.88 as the demonstrations provide different evidence for relevance; fixed-weight averaging keeps the normalized weight constant.

We regress the models' normalized weights on π_n, pooling $n = 1$, 2, and 4 at the stated 50% (269 replies per model). With the baseline prior stated, the slopes are 0.96 [0.84, 1.04] for GPT-6 Sol and 0.98 [0.82, 1.15] for Haiku 5.5, compared with 0.01 [−0.09, 0.11] and 0.22 [−0.01, 0.45] under the open-alternative design's warning. The 95% intervals resample the 153 tasks. The corresponding intercepts with the prior stated are 0.01 and 0.03, and R^2 is 0.57 and 0.33.

The agreement also holds when the demonstrations count against the description. Among the 23 replies with π_n below 0.5 (mean 0.33), the mean normalized weights are 0.30 and 0.38 for the two models with the baseline prior stated, compared with 0.48 and 0.50 under the open-alternative design's warning. Both models therefore vary their description weights across tasks in approximately the amount predicted by Bayesian model averaging.

Median ESS uses each model's baseline curve from the runs of Section 5.2. At the stated 50%, fixed-weight averaging gives 4.4 and the Bayesian reference gives 5.7. Stating the baseline prior raises ESS from 4.2 to 5.4 for GPT-6 Sol and from 3.8 to 4.2 for Haiku 5.5. At the stated 10%, it changes from 29.7 to 25.0 for GPT-6 Sol and from 32.6 to 15.1 for Haiku 5.5, compared with 27.5 for the reference. The two models' fitted weights agree within 0.01 in this condition (Table 12). Haiku's lower ESS therefore does not reflect lower average reliance on the description; it comes from a minority of replies far from the reference, some of which repeat a demonstration value. We also ran two further configurations with the baseline prior stated. Haiku 5.5 at low effort also updates (slope 0.93), with median ESS 2.7 at the stated 50%, against 5.5 under the open-alternative design's warning. Opus 4.6 without reasoning does not (slope −0.18), with median ESS 3.8 under both warnings.

Table 12. **Stating the baseline prior changes description weights.** Fitted coefficients on the $c = 36$ prior mean for the first 200 tasks, as in Table 11. "Unstated" denotes the open-alternative design's warning, "the mean is unknown"; "stated" specifies that the mean is otherwise uniform over 100 to 999, as the reference assumes. GPT-6 Sol uses low effort and Haiku 5.5 uses medium effort.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol, unstated	0.49	0.46	0.40	0.90	0.86	0.71
GPT-6 Sol, stated	0.58	0.59	0.59	0.88	0.83	0.80
Haiku 5.5, unstated	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, stated	0.62	0.60	0.63	0.89	0.84	0.80

C.7. Wording with Reasoning Enabled

Because reasoning brings description weights close to the reference under the original wording, we test whether it does so under the alternative wordings as well. We evaluate the three alternative wordings with Haiku 5.5 at medium effort, for the three relevant descriptions at $n = 1$, 2, and 4 on the first 200 tasks; all 5,400 replies contain a number. With "An earlier estimate put its mean at m, with a standard error of τ", at least 71% of replies in each cell include reasoning output; 99% of these lie within 2 of the reference prediction, the tolerance of Table 10. Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", reasoning output is absent from 4 to 36% and 2 to 74% of replies per cell, respectively. Even among replies with reasoning output, 71% and 91% repeat the prior mean, and 43% and 32% lie within 2 of the reference prediction; the two groups overlap, because repeating the prior mean can lie within 2 of the reference when the description is precise. At $c = 1$ and $n = 4$, its description weights are 0.72 and 0.93, compared with reference weight 0.20.

All wordings provide a value m and a scale τ, but "give or take" and "$\pm$" do not identify τ as a prior standard deviation or specify a distribution for the mean. The comparison varies the explicitness of the probabilistic specification as well as its wording. A model that combines a description with demonstrations correctly when the prompt states a distribution therefore need not do so when the prompt does not.

C.8. Descriptions in Words: Sorting Messages

The numerical descriptions state a distribution over the task, whereas descriptions in practice usually state constraints in words that the model must interpret. We therefore design a message-sorting experiment with such descriptions that still admits an exact Bayes-optimal predictor. As in Bayesian models of rule-based concept learning (Goodman et al., 2008), the hidden task is one of a finite set of rules, here 60 sorting rules under a uniform prior. A note constrains the rule in words: a broad note leaves 20 compatible rules, and a specific note leaves 2. With reasoning, Haiku 5.5 and GPT-6 Sol obtain ESS 9.9 and 9.5 for the specific note at $n = 2$, compared with 11.4 for the reference, and when the note may be irrelevant, most of their predictions are closer to the Bayesian reference than to fixed-weight averaging.

Task. A message has six binary properties: its sender is outside the organization; it requests a payment; it concerns travel; it requests computer access; it has a deadline within a day; and it contains personal contact information. Each message states every property as present or absent, using one of three phrasings for each value, such as "I'm a contractor from another company." or "I'm on the sales team here." Properties are sampled independently with probability one half, phrasings are sampled uniformly, and sentences appear in random order.

A rule uses exactly two properties and requires each to be present or absent. A message goes into folder A when both requirements hold, and into B otherwise. There are $\binom{6}{2}2^2 = 60$ such rules. The prompt describes this family, calling the rule the company's filing policy, and states that it was chosen uniformly from the family. It then provides a note, n previously sorted messages with their folders, and four new messages. The model is asked for each new message's probability of going into A, as a number between 0 and 1, on four lines from "1: <probability>" to "4: <probability>".

Notes. A broad note reads, for example, "Policy note: whether it concerns travel is one of the two properties the policy uses," leaving 20 possible rules. A specific note reads "Policy note: to go into A, a message must request a payment. The other property the policy uses is whether it concerns travel," leaving 2. These notes state constraints in words; the model must determine which rules satisfy them.

The unreliable note is a specific note with an added sentence: "This note has a 50% chance of being wrong, in which case it says nothing about the policy, which is then any of the 60, equally likely." In the generator, the note is relevant with probability one half and otherwise describes an independently drawn rule. The note is irrelevant in 92 of the 200 tasks. An independently drawn note can agree with the actual rule by chance, so we distinguish *relevance*, the latent component that generated the note, from *compatibility*, whether a rule satisfies it.

Reference and ESS. For a note asserted to be relevant, the reference predicts the fraction of rules consistent with the note and the sorted messages that assign the new message to A. For an unreliable note, it averages this prediction with the prediction from rules consistent with the sorted messages alone, weighting the two by the posterior relevance probability. The prior relevance probability is 0.5; the posterior probability is obtained using the likelihood of the demonstrations under each component. As a check on our implementation, on $3{,}000$ simulated tasks the reference's predictive probabilities are calibrated in every condition, and its posterior probability that the unreliable note is relevant matches the frequency of relevance: where it exceeds 0.7, it averages 0.80, and the note is relevant in 80% of cases.

The loss is the Brier score, the squared error between the predicted probability of A and the true folder indicator, averaged over the four query messages. Without demonstrations, a broad note leaves the prediction unchanged: for any message, exactly a quarter of the 20 rules compatible with the note send it to A, the same fraction as among all 60, so the broad note's ESS at $n = 0$ is zero. ESS follows Definition 1, using each learner's baseline curve, its loss without a note, at $n = 0$, 2, 4, 6, 8, 12, 16, 20, 24, and 32, with linear interpolation in n. On simulated tasks, this grid changes the reference ESS by at most 0.6 relative to evaluation at every integer demonstration count.

Runs. We use 200 tasks, each defined by a sampled rule, and 19 conditions: no note at the ten demonstration counts above, and each note at $n = 0$, 2, and 4. This gives $3{,}800$ prompts per configuration. Haiku 5.5 runs at medium effort; GPT-6 Sol runs without reasoning and at low effort. We leave out Opus 4.6, because in a pilot it still wrote out its working on this task with reasoning disabled, so its two settings would not differ only in reasoning. The command-line agents are set up as in Appendix C.4, with no tools, and no reply calls a tool. To check that GPT-6 Sol without reasoning reads the messages correctly, we asked it to state each property of 96 messages; it was correct in 97 to 100% of cases for each property; all three errors concern one phrasing, "I'm not traveling for this." Fourteen replies from GPT-6 Sol at low effort contain no probability and are excluded; one Haiku 5.5 reply omits one of its four probabilities, and we score its other three. Every Haiku 5.5 reply includes reasoning output, with a median of $3{,}008$ reasoning tokens. For GPT-6 Sol at low effort, 91% of replies include reasoning output, with a median of 142 reasoning tokens.

Results. Table 13 reports each note's ESS. With reasoning, both models are within 2 demonstrations of the reference in every condition except GPT-6 Sol's specific note at $n = 4$: its ESS is 26.1, compared with 14.8 for the reference. Its loss with this note, 0.008, is close to the reference's 0.007, but its baseline curve flattens beyond $n = 16$, with loss 0.022 at $n = 20$, compared with 0.004 for the reference. It consequently needs more demonstrations on its own baseline curve to match the note's error reduction.

For the unreliable note, we compare predictions with the Bayesian reference and with fixed-weight averaging, which averages the two component predictions using their prior weights. Here it retains the stated 50% weight unless the sorted messages rule out every rule compatible with the note; in that case, it assigns zero weight to the relevant component. We use query messages on which the two predictions differ by at least 0.05: about 150 at $n = 2$ and 200 at $n = 4$, out of 800. The share of predictions closer to the reference is 90% at $n = 2$ and 99% at $n = 4$ for Haiku 5.5; 77% and 87% for GPT-6 Sol at low effort; and 45% at both counts for GPT-6 Sol without reasoning.

Without reasoning, GPT-6 Sol learns slowly from demonstrations. Its loss without a note is still 0.078 at $n = 32$, compared with less than 0.0005 for the reference. Measured against this weaker baseline, a note can have an ESS as large as the reference's or larger. For example, the specific note has ESS 13.5 at $n = 2$, although the model's loss with that note is 0.130, compared with 0.027 for the reference. This illustrates why learner-specific ESS must be interpreted alongside absolute prediction error.

C.9. Complete Tables

Tables 14–16 report ESS and description weights at every evaluated demonstration count for the $2{,}000$-task run without reasoning.

Table 13. **ESS of notes in the message-sorting experiment.** Results use 200 tasks and each learner's own baseline curve, measured without a note. "≤ 0" means that the note did not lower loss; "> 28" means that the baseline curve does not reach the loss with the note by $n = 32$. On 6,000 simulated tasks, the reference ESS is 9.6 and 13.2 for the specific note at $n = 2$ and 4, and 2.5 and 2.9 for the unreliable note. Models are compared with the reference on the same 200 tasks. Paired 95% bootstrap intervals over tasks (1,000 replicates) for the specific note at $n = 2$ are: reference [9.7, 12.9], Haiku 5.5 [8.7, 11.5], GPT-6 Sol at low effort [7.8, 12.7], and GPT-6 Sol without reasoning [3.7, 21.9].

Note	n	Reference	Haiku 5.5 medium	GPT-6 Sol low	GPT-6 Sol no reasoning
Broad	0	0.0	0.0	≤ 0	≤ 0
	2	1.4	1.3	1.3	0.7
	4	1.8	1.6	1.7	8.6
Specific	0	3.7	3.6	3.3	3.7
	2	11.4	9.9	9.5	13.5
	4	14.8	14.5	26.1	> 28
Unreliable	0	1.3	1.3	1.6	1.7
	2	3.0	2.6	3.2	1.7
	4	5.0	4.9	5.1	2.7

Table 14. **ESS by description condition and demonstration count: distribution readout.** Results cover the Bayesian reference and the three distribution-readout models in the 2,000-task run without reasoning. "Number" is the numeric response rate averaged over n. "10% wrong" and "50% wrong" denote stated irrelevance probabilities. The censoring marker $\leq$ marks error above the first baseline point; $> 64 - n$ means that the error with the description lies below the entire baseline curve through 64 demonstrations.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Reference	$\sigma = 60, c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60, c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60, c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60, c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60, c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30, c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30, c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30, c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60, c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60, c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60, c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60, c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60, c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60, c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60, c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60, c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60, c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60, c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60, c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30, c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30, c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30, c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60, c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60, c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60, c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60, c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma = 60, c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma = 60, c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma = 60, c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma = 60, c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.5	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30, c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30, c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30, c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	> 32	> 0
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	0.85	≤ 0.0	≤ -1.0	-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60, c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	0.86	≤ 0.0	≤ -1.0	-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60, c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	0.85	≤ 0.0	≤ -1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60, c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	0.86	≤ 0.0	≤ -1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60, c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60, c = 1$, irrelevant	0.89	≤ 0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60, c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60, c = 4$, irrelevant	0.90	≤ 0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60, c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant	0.90	≤ 0.0	≤ -1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30, c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30, c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30, c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60, c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	≤ -1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60, c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60, c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table 15. **ESS by description condition and demonstration count: written readout.** Results cover the three written-readout models in the 2,000-task run without reasoning. Replies without a qualifying number are excluded. Columns and reference values follow Table 14.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Qwen3.5 9B (written)	$\sigma = 60, c = 1$, relevant	1.00	0.2	0.3	−0.9	−2.8	−9.3	−24	−51
	$\sigma = 60, c = 1$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.7	−4.8	−9.7	−23	−49
	$\sigma = 60, c = 4$, relevant	1.00	3.9	2.7	1.3	−0.7	−6.2	−18	−44
	$\sigma = 60, c = 4$, irrelevant	1.00	≤ 0.0	≤ -1.0	−1.7	−3.8	−9.4	−24	−49
	$\sigma = 60, c = 36$, relevant	1.00	6.5	4.0	2.8	6.9	17	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.8	−5.1	−10	−24	−51
	$\sigma = 30, c = 1$, relevant	1.00	≤ 0.0	0.2	−0.7	−2.5	−8.7	−22	−49
	$\sigma = 30, c = 4$, relevant	1.00	3.4	2.7	1.5	−0.3	−4.5	−14	−40
	$\sigma = 30, c = 36$, relevant	1.00	4.7	4.3	3.0	9.2	14	> 32	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	1.00	3.8	−0.4	−0.7	−1.8	−8.5	−21	−47
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−4.9	−10	−24	−50
	$\sigma = 60, c = 36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	−0.3	−3.3	−4.2	−6.9
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−5.2	−10	−25	−53
	$\sigma = 60, c = 4$, relevant, 50% wrong	1.00	4.5	−0.1	−0.4	−1.9	−9.0	−23	−50
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−6.2	−12	−26	−54
	$\sigma = 60, c = 36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	−0.1	−4.0	−4.2	−12
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−7.0	−13	−27	−56
Gemma 4 12B (written)	$\sigma = 60, c = 1$, relevant	1.00	0.0	0.0	−0.1	−0.6	−3.2	−5.4	−21
	$\sigma = 60, c = 1$, irrelevant	1.00	≤ 0.0	−0.4	−1.3	−1.3	−4.4	−6.8	−24
	$\sigma = 60, c = 4$, relevant	1.00	0.3	0.5	1.4	1.1	−0.6	−2.7	−13
	$\sigma = 60, c = 4$, irrelevant	1.00	≤ 0.0	−0.4	−1.1	−1.2	−4.1	−6.9	−23
	$\sigma = 60, c = 36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	> 0
	$\sigma = 60, c = 36$, irrelevant	1.00	≤ 0.0	−0.8	−1.3	−1.3	−4.9	−7.9	−26
	$\sigma = 30, c = 1$, relevant	1.00	≤ 0.0	−0.2	−0.1	−0.5	−3.2	−7.9	−27
	$\sigma = 30, c = 4$, relevant	1.00	0.7	0.8	2.1	1.5	−0.2	−4.6	−18
	$\sigma = 30, c = 36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	−1.8	−8.9
	$\sigma = 60, c = 4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	−1.0	−3.1	−7.3
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.4	−0.5	−2.2	−5.4	−13
	$\sigma = 60, c = 36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	−0.6	−1.3
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.5	−0.5	−2.5	−5.8	−15
	$\sigma = 60, c = 4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	−0.1	−1.7	−4.5
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.4	−0.4	−1.3	−4.1	−9.6
	$\sigma = 60, c = 36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.6	−0.4	−1.9	−4.8	−13
Gemma 4 31B (written)	$\sigma = 60, c = 1$, relevant	1.00	0.1	0.0	−0.1	−0.5	−1.4	−5.1	−13
	$\sigma = 60, c = 1$, irrelevant	1.00	≤ 0.0	−0.1	−0.1	−0.5	−1.0	−4.5	−11
	$\sigma = 60, c = 4$, relevant	1.00	0.8	0.1	0.2	−0.5	−1.0	−4.3	−12
	$\sigma = 60, c = 4$, irrelevant	1.00	≤ 0.0	−0.2	−0.2	−0.4	−1.0	−4.4	−10
	$\sigma = 60, c = 36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	−2.9
	$\sigma = 60, c = 36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−0.8	−0.6	−1.1	−2.7	−4.8
	$\sigma = 30, c = 1$, relevant	1.00	≤ 0.0	0.0	−0.1	−0.7	−1.2	−3.0	−8.1
	$\sigma = 30, c = 4$, relevant	1.00	0.5	0.0	0.1	−0.5	−1.3	−3.4	−8.3
	$\sigma = 30, c = 36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	−0.4
	$\sigma = 60, c = 4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	−0.7	−1.9	−5.0	−15
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.2	−4.1	−9.5
	$\sigma = 60, c = 36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	−0.3	−1.4	−4.8	−12
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−0.9	−3.0	−6.2
	$\sigma = 60, c = 4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	−0.7	−1.7	−4.6	−17
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−1.2	−3.6	−12
	$\sigma = 60, c = 36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	−0.3	−1.5	−4.5	−15
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.0	−2.9	−8.5

Table 16. **Description weights by demonstration count: distribution readout.** Results cover the Bayesian reference and the three distribution-readout models in the 2,000-task run without reasoning. "Number" is defined as in Table 14; cells are italicized when the numeric response rate is below three quarters at the corresponding n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Reference	$\sigma=60, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60, c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60, c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60, c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60, c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60, c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60, c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60, c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60, c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60, c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60, c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60, c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30, c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30, c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30, c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60, c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60, c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60, c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60, c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60, c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60, c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60, c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60, c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60, c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30, c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30, c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30, c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60, c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma=60, c=1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma=60, c=1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma=60, c=4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma=60, c=4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma=60, c=36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma=60, c=36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma=30, c=1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma=30, c=4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma=30, c=36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma=60, c=36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma=60, c=4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma=60, c=36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01

D. Additional Related Work

Task descriptions and demonstrations. Weller et al. (2020) formalize learning from task descriptions by distinguishing the latent task, its natural-language description, and individual input–output examples. We use *task description* for information about the task supplied in the prompt, including a numeric description token in the meta-trained setting and probabilistic statements or notes about a sorting rule in the language-model experiments. A *demonstration* is an observed input–output example; in scalar mean estimation, it is an observation at the fixed input $x = 1$. Summaries distilled from demonstrations can serve as descriptions (Honda et al., 2025); our controlled experiments instead specify the joint distribution of task, description, and demonstrations.

Prior information and unreliable descriptions. The two-component construction has the form of the robust mixture priors of Bayesian clinical trials, which combine informative and baseline distributions and update their component weights after observing data (Schmidli et al., 2014). Our mixture is specified directly, rather than derived from a historical-data meta-analysis. Reliability p is the prior probability of the relevant component; π_n is its posterior probability. Averaging component predictions using π_n is Bayesian model averaging (Hoeting et al., 1999). In a robust mixture prior, the prior weight is fixed but the posterior weights still update; fixed-weight averaging, our experimental comparison, omits this update.

In our model, an irrelevant description is independent of the task and may happen to agree with it. Irrelevance is therefore a property of how the description was generated, whereas prior–data conflict concerns disagreement between prior information and observed data.

Specificity and description weights. Specificity takes a concrete form in each setting. Section 2 measures it by what a relevant description leaves unknown about the task: the smallest expected loss of a prediction made from the description's prior alone, which Grünwald & Dawid (2004) call the generalized entropy of that prior under the loss. Under squared error this is the prior's variance, which in the Gaussian model is τ^2 in each direction that the description constrains; a partial description constrains k of them. A message-sorting note leaves a uniform prior over the rules compatible with it, and we describe its specificity by their number, 20 for a broad note and 2 for a specific one; the logarithm of that number is the prior's entropy, the corresponding measure under log loss. These quantities do not share a scale across tasks; ESS expresses each in demonstrations. Our description weights follow Gaussian mean estimation, in which the posterior combines the prior mean and the sample mean with weights proportional to their precisions (Neuenschwander et al., 2020). Fitted and per-response weights summarize model predictions; their normalized values can be compared with posterior component probabilities but need not be probabilities.

Effective sample size and predictive comparisons. Prior ESS has several definitions, including matching local information (Morita et al., 2008), matching expected posterior loss (Reimherr et al., 2021), and quantifying sample savings at a target estimation accuracy (Wiesenfarth & Calderazzo, 2020). Matching information requires an explicit prior. A network or a language model has prior knowledge from training, but only implicitly, in its parameters; its prediction error, by contrast, can be measured with and without the description. Our definition therefore follows the error-matching approach, but it evaluates population prediction error, allows a non-Bayesian learner, and depends on the demonstration count. A learner without a description still has prior knowledge, from the baseline prior or from training, and ESS measures what the description adds to it, so we measure ESS against each learner's own baseline curve. For a learner whose baseline curve falls slowly, the same error reduction is worth more demonstrations, so a large ESS need not imply low absolute error (Table 9). A negative ESS means that the description raises error. The word *predictive* specifies what is matched; it does not assert the additive predictive-consistency property studied by Neuenschwander et al. (2020).

Two predictive comparisons are especially close. Le Scao & Rush (2021) average horizontal data advantages over an accuracy range in prompt-based fine-tuning; we evaluate a pointwise difference in demonstration counts. Selby et al. (2025) compare priors elicited from language models with data-based predictors by the observations needed to match predictive loss; we study descriptions supplied to a learner and how their benefit changes after demonstrations. Other work compares instructions or prompts with examples. Beyond ICL, an instruction can replace up to 100 execution examples in the fine-tuning settings studied by Cook et al. (2026), and Wenliang et al. (2026) compare optimized prompts with demonstrations in terms of their length and effect.

Demonstrations, text, and probabilistic prediction. Other work examines semantic priors and flipped labels (Wei et al., 2023), weak dependence on demonstration labels (Min et al., 2022), Bayesian scaling laws (Arora et al., 2025), and Bayesian reference curves for in-context function learning (Akata et al., 2026). Text supplied alongside numbers can change numerical predictions (Requeima et al., 2024). Related work studies how language-model outputs compare with Bayesian inference and how training can improve probabilistic reasoning (Qiu et al., 2026); He et al. (2025) examine Bayesian consistency during iterative reasoning on open-ended questions. Our numerical experiments state uncertainty explicitly, allowing comparison with a specified Bayesian reference as precision and relevance probability vary. The message-sorting experiment extends this comparison to constraints stated in words on a finite family of classification rules.

Scope of the results. The $d + 1$ term of Proposition 1 arises from random regression inputs; it is absent when every input is the same, as in our mean-estimation task (Appendix C.2). For language models, the revision of a stated chance of irrelevance carries over to a description that states a rule in words (Appendix C.8). Agreement with the reference does not carry over to informal wording: when the prompt gives the prior mean "give or take" an amount, most of Haiku 5.5's replies with reasoning repeat the prior mean (Appendix C.7).